VR Anatomy Learning: Kidney Visualization

Areeba Amin
(01-134222-028)

Atika Abid
(01-134222-032)

Supervisor: Dr. Muhammad Asfand E Yar

Bachelor of Science in Computer Science

Department of Computer Science
Bahria University, Islamabad

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Certificate

We accept the work contained in the report titled **VR Anatomy Learning: Kidney Visualization**, written by **Areeba Amin and Atika Abid** as a confirmation to the required standard for the partial fulfillment of the degree of Bachelor of Science in Computer Science.

Approved by

Supervisor: Dr. Muhammad Asfand E Yar (Senior Assistant Professor)

____

Internal Examiner:

____

External Examiner:

____

Project Coordinator: Dr. Asim Ali

____

Head of Department: Dr. Khuram Ehsan

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Abstract

Our project introduces an interactive and engaging VR (Virtual Reality) based anatomy learning system that is aimed at improving the medical education using 3D visualization and functional simulations. In contrast to the conventional methods of teaching which rely on cadavers,textbooks our system provides a dynamic virtual environment, in which students are able to study a 3D model of the kidney in detail, interact with it, and examine its internal structure.By combining zoom-in functionality,mesh separation,and animated visualizations of organ functionality,the platform turns anatomy learning into a captivating and a practical learning experience. Research in medical learning education emphasizes that VR-based learning is more engaging and understanding as compared to the usual cadaver-based or textbook approaches. Its interactive quality enables learners to learn complex structures in ways that are not possible in static or physical environments. The system focuses on Human kidney, providing a 3D model with separate meshes of its internal structures each labelled for teaching purposes. There are also functional animations that is accomplished by particle systems that enable learners to learn not just the anatomy but also the physiology of the organ. The nephron is also a part of the system,which is the functional unit of the kidney and simulation of diseases like kidney stones and polycystic kidney disease, contributing to structural and clinical insights.Beyond improving knowledge, VR also has other advantages like increasing the motivation of the learners,enhancing focus, and creating a more productive and engaging learning experience.

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Areeba Amin,Atika Abid
Bahria University
E-8, Islamabad, Pakistan
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Acronyms and Abbreviations

VR	Virtual Reality
XR	Extended Reality
UI	User Interface
GUI	Graphical User Interface
3D	Three-Dimensional
HMD	Head-Mounted Display
SDK	Software Development Kit
API	Application Programming Interface
IDE	Integrated Development Environment
FBX	Filmbox (3D Model File Format)
FPS	Frames Per Second
C#	Programming Language used in Unity
CPU	Central Processing Unit
GPU	Graphics Processing Unit
PKD	Polycystic Kidney Disease
UV	Texture Mapping Coordinates (U-V Mapping)
AR	Augmented Reality
AI	Artificial Intelligence
OS	Operating System
MP3	Audio File Format (MPEG Layer-3)
HLD	High-Level Design
LLD	Low-Level Design
SDLC	Software Development Life Cycle
VR UI	Virtual Reality User Interface
XR Toolkit	Extended Reality Interaction Toolkit

CHAPTER 1

INTRODUCTION

1.1 Project Overview

The invention of the virtual reality (VR) technology has revolutionized the manner. Teaching and learning take place on complex subjects, especially on the sphere of medical and biological sciences. Anatomy of human beings is a basic subject to be studied among the medical students; but, conventional learning tools like textbooks, 2-dimensional pictures and fixed models do not give a clear picture about the three dimensional structure of the spatial association of organs and the human body.These challenges will be addressed in the project, VR Anatomy Learning: Kidney Visualization through offering an interactive learning environment through virtual reality.

It enables users to navigate through the detailed 3D models of the kidney and nephron, comprehend the inner structures with the help of interactive features, and observe the processes within the body with the help of animations. Disease simulations like Polycystic Kidney Disease and kidney stones are also part of the system to offer real world medical knowledge. The application is created using Unity, Blender, and the XR Interaction Toolkit and runs on the Meta Quest 2 headset and provides an engaging and immersive educational experience to students.

Research shows that virtually every participant enjoyed the VR experience and making it educationally useful with a 95% confirmation of the ease of use of the system, persuasive immersion, and straightforward interaction [1].

1.2 Problem Description

The Traditional ways of learning anatomy are based on the use of static materials that do not represent the dimension of the organs.Students have a tendency to fail in grasping the location and anatomy of the kidney since it is located very deep within the human body.These limitations can be overcome with an immersive VR-based learning system enabling the study of the layers of the anatomy interactively and learn the inside organ structure. [2].

Traditional methods of anatomy teaching involve the use of static models that do not represent the dimension of organs

1.3 Project Objectives

The main goal of this project is to create an immersive virtual reality application for kidney anatomy learning. The specific objectives to develop a VR system are:

1. Build a structured system architecture with the help of Unity and the XR Interaction Toolkit which enables smooth transition among 3D models(Kidney,Nephron).
2. Implement a layer-by-layer visualization method in which students are able to interactively separate or examine kidney meshes to inspect internal physiological structures in a 3D spatial context.
3. Use the Unity Animator and Particle Systems to provide functionalities of the kidney along with 2 kidney diseases.
4. Add a multimodal learning strategy with the synchronization of 3D visual labels with an audioManager.
5. Develop a risk-free, self-paced virtual environment that lessens the reliance on physical cadavers or static 2D textbook diagrams.

1.4 Project Scope

This project aims at developing an interactive world in VR to facilitate learning among the students about the human kidney. The system starts by a naviagtion bar. Using a navigation menu the user is able to change learning modes using five large buttons. The ShowKidney and Show Nephron buttons enable the user to view high quality 3D models. They can zoom in on and seperate the parts to study in detail. The Show Diseases button allows the student to investigate two particular conditions: Polycystic Kidney Disease (PKD) and Kidney Stones. In addition, the Anatomical Position and Kidney Animations buttons give us a glimpse of the position of the kidney within the body and the functioning of the organ. This project is created in Unity and the XR Interaction Toolkit, on the Meta Quest headset, and it is strictly for education, not to treat real patients.

1.5 Solution Application Areas

The VR-based anatomy learning system is of great importance in various domains:

1. **Medical Education:** The system is directly involved in medical and allied health students by way of interacting anatomy and physiology. Unlike students that practice on cadavers, or stationary textbooks, etc. students can explore 3D model of a kidney and it's functionality that leads to increased retention and conceptual clarity [3].
2. **Training & Skill Development:** It may be an additional aide of medical colleges,nursing education and medical training schools to enhance understanding,without the moral and practical limitations of using cadavers.
3. **E-Learning & Remote Education:** VR-based modules allow students to learn from anywhere.It makes high quality anatomy education accessible to learners in

remote or resource limited areas.

4. **Healthcare Industry:** It is possible to apply the healthcare system to hospitals and clinics in educating patients, helping people to be aware of their health issues(e.g. Kidney Diseases) through the demonstrations created visually of how the organs work.

5. **Research & Development:** This system can be served as a foundation in the future other organ expansion, training surgical, and advanced disease modeling.

1.6 Risks Involved

VR-based development of anatomy learning system as any other software development project entails certain risks. The risks can include the process of development, performance of the system, or user experience.It is expected that the project is based on the VR technology and 3D Models that can lead to challenges, including technical limitations, flexibility of the user and problems in Reliability and Durability.However,the risks may be minimized with proper design, testing and professional instructions.

1.6.1 Technical Challenges

Among the biggest risks, there is a risk associated with technical issues. VR development involves work including Unity, 3D models and VR hardware that can be tricky and time-intensive. The 3D models, which are being built in large scale or detailed models may result in performance issues such as lag or slow rendering. In order to reduce this risk, optimized models will be utilized and the system will expand over time with periodic testing and checking.

1.6.2 User Adoption and Usability

The other possible risk is the adoption by the user since not all the users would be familiar with virtual reality systems. The usage of VR may be an issue for the new users. Some users can also experience motion discomfort. The system will have easy controls, easy navigation in order to manage this risk. With the help of user testing and feedback,the ease of use and the general experience will be enhanced.

1.6.3 Content Accuracy and Educational Reliability

In the current-day world, the accuracy of the presented content and its educational reliability is very important. As the system is applied in education,any wrong or ambiguous anatomical information may lead to a serious risk. Inaccurate models or animations may create confusion amongst students. To avoid this good anatomy models will be picked and the content will be checked with the assistance of instructors or medical experts to ensure Educational accuracy.

1.6.4 Security and Data Safety

Though it is not a system that keeps any sensitive personal or medical information, but there are still some basic security risks such as, the system being unauthorisedly accessed or changed by accident. These problems may have an impact on the stability of the system. In order to reduce this risk, the application will be developed using common software development practices and trusted tools in a controlled environment

1.7 Summary

The project VR Anatomy Learning: Kidney Visualization is an interactive VR environment and allows learning about the human body, in particular, about the kidney. The users are able to separate meshes and can view the labeling and description of the parts and take a look at functional animations. It enhances the medical education, development of skills and distance learning, conquering the disadvantages of textbooks and predetermined prototypes. Optimization, testing and expert validation shall be used to address technical issues,adaptation by users and accuracy of the contents.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The fast development of digital technologies has impacted the world of to a great extent, particularly in the medical and biological sciences.The traditional method of studying anatomy is through books, posters, plastic models and cadaver dissection and the drawbacks of this method are cost, ethical considerations and lack of resource availability. Through virtual reality, it is possible to use an interactive mode of dealing with anatomical models and learners are able to study the internal structures dynamically and in different viewpoints. Simulation learning has been shown to have an environments enhance participation, memorization and spatial awareness. The VR Anatomy Learning:Kidney Visualization project is based on the above innovations which include a level of exploration which is camera based and adds insights to the internal anatomy structure. [4].

2.2 Related Applications

There are several digital and virtual reality-based applications developed to facilitate anatomy learning as an indication of the interest in immersive learning. But there are some limitations of the systems that the proposed system seeks to overcome. 3D anatomy programs are providing models that can be zoomed, isolated and rotated. Although these systems are better rendered in external appearances of organs in contrast to the two-dimensional diagrams, interaction is limited to external features of organs. This inhibits students' opportunity for the internal anatomy, which is essential in understanding spatial relationship of human body. Virtual dissection applications go a step further to allow the user to dissect body parts virtually. Such apps are normally dependent on manual layer switching or object selection which could hinder learning. Such ways of communication to beginners may be confusing, as they might not be familiar with the anatomy structures.

2.2.1 Complete Anatomy(3D4Medical)

Complete Anatomy (3D4Medical) is an application where the system offers high 3D quality anatomical models but does not have dynamic functionalities. Primarily the models are retained static and there are few physiological simulations to be interacted with by students. Our project is one way of filling this loophole where dynamic functional animations are introduced,blood filtration and urine formation and interactive mesh separation which enables learners to investigate the organs individually. [5].

Table 2.1: Comparison of Related Applications

Application	**Weaknesses**	**Project Solution**
Complete Anatomy (3D4Medical)	DMainly Static 3D models without dynamic functionality. Limited interactive physiological simulations.	Dynamic functional animations indicating the blood filtration and urine formation. Interactive Mesh separation.
Human Anatomy VR	Basic visualization without functional understanding. no physiological systems, process demonstrations.	Interactive using zoom and mesh separation.Active learning through hands-on exploration. Physiological process demonstrations.
Cadaver-based learning	Limited availability and high maintenance costs. No physiological process demonstrations.	No ethical or health issues. Visualization of living physiological processes. Reusable satisfied with stable quality.
3D Organon XR	Provides a very broad anatomy but does not have details functional physiological animations (emphasis is placed more on structural anatomy). High costs of licensing also restrict accessibility for students.	Focuses on one organ (kidney) and in depth interactive functional animations.

2.2.2 Human Anatomy VR

Human Anatomy VR offers an easy visualization, and does not enable functional understanding of organ system. It is not very interactive and fails to show physiological processes. Our project in the other hand is concentrated on the high interactivity, zoom, mesh separation, and functional process demonstrations. This fosters active learning and more engagement [6].

2.2.3 Cadaver-based learning

The use of traditional cadavers is still a traditional approach to medical education, but it is very limited.Cadavers are costly, not easily accessible and are challenging due to ethical concerns. Moreover, cadaver dissections are unable to illustrate living physiological processes. Our solution will remove these barriers by providing reusable VR content that consistently is of good quality and enables visualization of real time physiological functions.

2.2.4 3D Organon XR

The 3D Organon XR is a highly established platform, which offers a wide range of structure of anatomy, it is mainly focused on visualization, not functionality. It lacks active interaction and detailed functional simulations e.g. organ processes.Our system is distinguished on the basis of anatomical visualization integration with physiological processes, interactive simulations,which helps structural and functional studying in a single VR space [7].

2.3 Tools and Technologies Used

The proposed VR-based anatomy learning system uses various tech-based tools and technologies to allow practical and immersive learning. These technologies work together to make it easier to visualise, touch and experience the real anatomy and their functionality.

- **Virtual Reality (VR) Technology:** Virtual Reality allows users to experience a three-dimensional space where they will be able to experiment with the kidney interactively. It helps the students to study the complicated anatomy.
- **Unity 3D Game Engine:** Unity is the primary development platform of creating the VR world. It deals with Live rendering, interaction management, and control of animation in the system.
- **Unity XR / OpenXR Framework:** It is a framework that is employed in the VR headset integration and processes user input either via controllers or by eye-tracking. It is used to improve performance and accuracy with VR.
- **Blender (3D Modeling Tool):**Blender is a tool used in editing, optimization and separation of 3D meshes of the skeleton and kidney models. It helps in the improvement of performance and accuracy to be used in VR.
- **Animation and Particle Systems:** The animation tools and particle systems in Unity are used to simulate the renal mechanisms such as blood filtration and urine production,simplifying physiological processes.
- **C# Programming Language:** C# is a programming language that is used to type the scripts of controlling the zoom, organ selection, mesh separation, and animation play interactions.

These technologies combined provide a smooth, interactive and effective virtual learning experience in anatomy.

2.4 Related Research Studies

2.4.1 Virtual reality and the transformation of medical education

The virtual reality helps in making medical education more efficient since it provides immersive and replicable learning. It has been found in the research [8] that the VR will decrease the high cost and low accessibility of conventional simulation based training and enhance the knowledge gain through experiential learning.This will be used to make better medical education at a lower cost and scale.

2.4.2 The effectiveness of virtual reality-based technology on anatomy teaching

The study [9] did a meta-analysis that found out that virtual reality based teaching is better in performance in anatomy than the traditional one. The research shows that VR is a more effective tool for teaching learners about the anatomy concepts that are difficult to communicate in books or on slides.The results demonstrate that VR can be used as an effective method of enhancing the knowledge of the anatomy and learning outcomes.

2.4.3 Systematic analysis of anatomy virtual reality (VR) apps for advanced education and further applications

The study [10] has analyzed the existing anatomy-based VR applications and found out that the existing VR technologies offer more than a visual representation. Some of the they list interactive 3D exploration and as characteristics in their study, greater learning abilities to enable students to learn complicated spatial relationships.The research validates the utilization of VR as a harmless and reproducible learning setting for anatomy education.

2.4.4 Impact of VR on Learning Experience compared to a Paper based Approach

The experiment [11] entailed a comparison of VR based learning, and traditional paper based learning and concluded that VR is much better in motivating the learners, getting them involved and also how they learn best in general.Despite resemblance in immediate learning outcomes, VR created more interest and positive learning emotions in students. This demonstrates that the immersive learning conditions may develop the long-term involvement as well as the willingness to learn.

2.4.5 Can virtual reality improve traditional anatomy education programmes? A mixed-methods study on the use of a 3D skull model

The study [12] made a comparison of a VR skull model with cadaver specimens and 2D atlases, discovering that VR is as effective in learning anatomy as the conventional techniques. While there were no significant differences in test scores between groups, VR intervention had a significant positive impact.

2.4.6 Gaps Identified

Despite the fact that previous studies prove the positive effect of VR in medical and learning anatomy, it has its limitations yet to be applied, there are gaps which limits the generalizability and practical implementation of these findings.

1. **Lack of Curriculum Integration** The majority of existing VR anatomy applications are not yet a part of official medical training. They are often used as a structured systems of learning more in line with supplementary or experimental tools, learning outcomes, course assessments, and objectives. This restricts their usage in academic institutions.

2. **Limited Long Term Learning Evaluation** Current studies primarily focus on immediate learning outcomes after VR usage.There is insufficient evidence regarding long-term knowledge retention, conceptual understanding, and continuous learning

impact. Additionally, factors such as cybersickness and prolonged VR exposure are rarely evaluated.

3. **Focus on Structure over Function** Most available applications emphasize visual and structural anatomy rather than functional understanding.Physiological processes such as blood filtration and organ-level interactions are often missing or minimally represented, limiting functional learning.

4. **Engagement without Measured Learning Efficiency** Although VR systems significantly improve learner engagement and motivation, their impact on actual learning efficiency and academic performance remains unclear. The novelty of immersive environments may distract learners rather than support long-term educational outcomes.

These gaps indicate the need for VR based anatomy systems that not only provide immersive and engaging experiences but also ensure long-term knowledge retention, functional understanding, and integration into standard medical curriculum. The present study aims to address these gaps by focusing on interactive kidney exploration with functional animations.

2.5 Summary

In this chapter, the current anatomy learning systems and the related technologies were reviewed and investigate the area of medical education by use of VR. The review emphasized the fact that the existing applications are oriented towards the static visualization and lack interactive physical demonstrations of physiological processes.

Moreover, the essential gaps in the research were also revealed, such as weak functional learning, absence of long-term assessment, and poor medical integration.Such loopholes explain why the proposed VR-based anatomy learning system should be developed.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Existing Systems

The current learning systems of anatomy are based mainly on 3D visualization to aid in teaching the anatomy. The systems enable the user to rotate the models, enlarge or shrink the organs and other body parts, to see them better. Although these features help one develop a better sense of visual perception than the two dimensional learning techniques of the past, they are mostly restricted to surface level interaction [5].

Most of the current systems present organs as predefined models and provide minimal assistance in the dynamic processes which occur in the physiology. Consequently, the students are usually able to identify the body parts but fail to understand how processes can take place in real time. This difference between the structure and the functionality of the body reduces the effectiveness of the anatomy teachings particularly when it comes to organs such as the kidney which is complex.

Virtual dissection based applications are more interactive as they enable the users to remove digitally anatomical layers. Nevertheless, these systems mostly depend on layer switching manually or object selection which can disrupt the learning process. Such methods of interaction may be difficult to understand for the beginners, who may lack sufficient background knowledge about the anatomy organization. [6].

In spite of three dimensional visualization of anatomy offered by immersive VR platforms, which uses the head mounted displays, most of the systems do not attempt to preserve anatomical context in the human body but instead show individual organs in three dimensional form. Besides, the interaction mechanisms and hardware requirements may be tricky and restrict usability and accessibility to early stage learners.

Generally, the existing systems emphasize more on visual representation than on functional knowledge, lack user friendly layout in terms of navigation through the anatomical layers, and and do not easily incorporate the anatomy with the physiology.These restrictions underscore the necessity to have an anatomy learning system that is more immersive, interactive and even functionally rich.

3.2 Proposed Solution

The proposed Virtual Reality based Anatomy Learning System is designed to tackle the induced shortcomings of the existing anatomy teaching approaches. The proposed solution is likely to incorporate the structural anatomy with physiological processes in the virtual reality which is immersive and intuitive.

The system offers an interactive virtual reality (VR) experience of learning kidney anatomy in a more engaging and intuitive manner than the old ones. This system, in contrast to the traditional methods that work with the fixed models or diagrams in textbooks, enables users to investigate 3D anatomical structures in the VR environment. The users are able to navigate to various learning modules via a navigation menu thus having a guided and systematic learning experience.

The system is based on structural and functional knowledge of kidney. It has 3D model of the kidney and the nephron,which users can interact with by using VR controllers. The models can be handled closely through grab interaction, enabling the user to have a better view of the anatomical details and spatial relationships.

The system also has animations that can be used to improve the conceptual knowledge of relevant physiological processes like filtration and urine formation. Such animations are useful to reduce the gap between the theoretical and conceptual understanding of how the kidney works.

Moreover, the system offers visualization of the diseases, which enables users to learn about conditions like Polycystic Kidney Disease (PKD) and kidney stones. This will aid the learners in relating the anatomical knowledge in real-life medical situations.

Virtual reality technology enhances depth perception, immersion and interaction with the user. The interaction design is maintained minimal with grab and selection-based interactions that makes the system easy to operate.

3.3 Target Audience

The main audience of Virtual Reality-based Anatomy Learning System would consist of medical students, high school and college biology students, and any one who likes to learn anatomy. The system will accommodate the formal classroom learning as well as self-directed study which will be flexible to many educational setups.

The system is designed particularly to meet the needs of beginners having little or no background knowledge of anatomy. It will assist in simplifying complicated anatomy concepts.Delivering user-friendly navigation,detailed visualization and demonstrations of physiological processes and understanding. This is an effective teaching tool in the early steps of anatomy education as the design available is accessible to the learners without overwhelming them.

3.4 Customer Requirements

The major client of the proposed system is medical and allied health students, and the teachers who must have a good and interactive tool to teach human anatomy.The customer wants a system which will give the students a chance to study the kidney in virtual space of reality, look at the anatomical position, select and grab the kidney internal parts, and find clear labels and descriptions for each part.They also desire to view functional animations.Simple navigation with easy to use controls, and a reusable and ethical alternative to cadaver based learning. To meet the needs, we have designed a VR based learning system of anatomy with immersive,interactive, and modular learning, allowing the students to learn about the kidney in detail while supporting future expansion to other organs and functions.

3.5 Functional Requirements

Functional requirements explain the main features and functionality that the VR-Based kidney learning system must perform. These are the requirements that specify how the system should react to user interactions and and how different components of the application work together to provide an interactive learning experience. The functional requirements include visualization of the kidney and nephron models,disease demonstrationtions,visualization of anatomical position, and display of labels and descriptions. The following tables describe each functional requirements in detail.

Table 3.1: Functional Requirement FR-1: 3D Anatomical Model Construction

Field	**Description**
Functional Requirement ID	FR-1
Functional Requirement Name	Building 3D Anatomical Models
Inputs	The anatomical reference data, 3D mesh files generated in Blender, texture, and material properties.
Process	The system imports 3D anatomical models into Unity. The rendering pipeline allocates materials, initializes mesh transforms, prepares the models to be interacted with in VR.
Outputs	Fully rendered 3D anatomical models displayed in the VR environment
Description	The system builds up correct 3D anatomical models of the human body and kidney through bringing new assets and preparing them towards interactive VR visualization.

Table 3.2: Functional Requirement FR-2: VR Environment Navigation

Field	**Description**
Functional Requirement ID	FR-2
Functional Requirement Name	VR Environment Navigation
Inputs	Reads VR headset tracking data, controller joystick input, and gaze direction
Process	XR interaction modules process headset and controller input. Locomotion scripts update camera position and rotation while enforcing movement constraints.
Outputs	Updated camera position and orientation in the virtual environment
Description	The system enables users to move, rotate, and zoom within the VR environment to explore anatomical structures.

Table 3.3: Functional Requirement FR-3: Kidney Model Visualization

Field	**Description**
Functional Requirement ID	FR-3
Functional Requirement Name	Kidney Model Visualization
Inputs	User presses the "Show Kidney" button
Process	The system activates the 3D kidney model in the Unity scene and enables interaction scripts for exploring the organ.
Outputs	A complete 3D kidney model that is visible in the VR environment
Description	The system enables users to view the kidney in detail and start to investigate its internal structure.

Table 3.4: Functional Requirement FR-4: Kidney Component Separation

Field	**Description**
Functional Requirement ID	FR-4
Functional Requirement Name	Kidney Component Separation
Inputs	User interaction with the kidney model
Process	Unity scripts separate the kidney mesh into individual anatomical parts such as cortex, medulla, and renal pelvis using predefined transformations.
Outputs	Separated kidney components for detailed visualization
Description	The system allows the user to investigate internal kidney structure by seperating various parts of the kidney model

Table 3.5: Functional Requirement FR-5: Nephron Model Visualization

Field	**Description**
Functional Requirement ID	FR-5
Functional Requirement Name	Nephron Model Visualization
Inputs	User presses the "Show Nephron" button after activating the kidney model
Process	The system enables the nephron 3D model and activates interaction scripts that allow mesh separation and exploration.
Outputs	Detailed nephron model shown in the VR environment
Description	The system enables the user to learn about the nephron structure,which is the functional unit of the kidney.

Table 3.6: Functional Requirement FR-6: Anatomical Labels and Descriptions

Field	**Description**
Functional Requirement ID	FR-6
Functional Requirement Name	Anatomical Labels and Descriptions
Inputs	User interaction with kidney or nephron parts
Process	The system identifies the selected component and displays labels and descriptive educational text in the VR environment.
Outputs	Labels and descriptions of anatomical parts
Description	The system offers learning support by showing names and descriptions of structures of kidney and nephron.

Table 3.7: Functional Requirement FR-7: Kidney Function Animation

Field	Description
Functional Requirement ID	FR-7
Functional Requirement Name	Kidney Function Animation
Inputs	User presses the "Show Animation" button
Process	The animation controller plays a predefined animation describing kidney functions while synchronizing background audio narration.
Outputs	Animated visualization of kidney function with audio explanation
Description	The system demonstrates the process of the kidney through animation and voice narration to improve learning.

Table 3.8: Functional Requirement FR-8: Anatomical Position Visualization

Field	Description
Functional Requirement ID	FR-8
Functional Requirement Name	Anatomical Position Visualization
Inputs	User presses the "Anatomical Position" button
Process	The system displays a full human body model and after a short delay moves the user camera inside the body to show the kidney's anatomical position.
Outputs	Visualization of kidney position within the human body
Description	The system helps users understand the anatomical position of the kidney inside the human body.

Table 3.9: Functional Requirement FR-9: Kidney Disease Visualization

Field	Description
Functional Requirement ID	FR-9
Functional Requirement Name	Kidney Disease Visualization
Inputs	User presses the "Show Diseases" button and selects a disease option
Process	The system activates the selected diseased kidney model such as polycystic kidney or kidney stones and plays the corresponding audio explanation.
Outputs	Diseased kidney model displayed with audio description
Description	The system allows users to learn about kidney diseases by visualizing affected kidney models and listening to educational explanations.

3.6 Non-Functional Requirements

Non-functional requirements elaborate the quality attributes and performance characteristics of the system. These needs are concerned with the system performance rather than what the system does. They include aspects such as performance of systems,usability, reliability, security and compatibility with VR hardware. These requirements make sure that the system offers easy, interactive, and user friendly learning experience within the virtual reality environment.

Table 3.10: Non-Functional Requirements

Sr. No.	**Non-Functional Requirements**	**Category**
1	The system has to display 3D models and animation smoothly at a minimum of 60 FPS without lag.	Performance
2	The virtual reality interface should be user friendly with simple navigation and clear labels.	Usability
3	The system should not crash during organ exploration or interaction.	Reliability
4	Models, labels and animations of the kidney should be scientifically right and justified by the experts.	Accuracy
5	The system must be compatible with standard VR headsets and high performance PCs with GPU integrated.	Compatibility
6	The system must be able to accommodate the addition of new organs, layers, and functional animations.	Scalability
7	The system should be in a position to support left and right handed users and minimize motion discomfort in VR.	Accessibility

3.7 Use Case Diagrams and Descriptions

In system analysis, Use Case Diagram is used to identify, clarify, and organize system requirements. It shows the interaction of the Actor with the system. It provides a solution of the tasks that a certain actor can especially perform with the system.

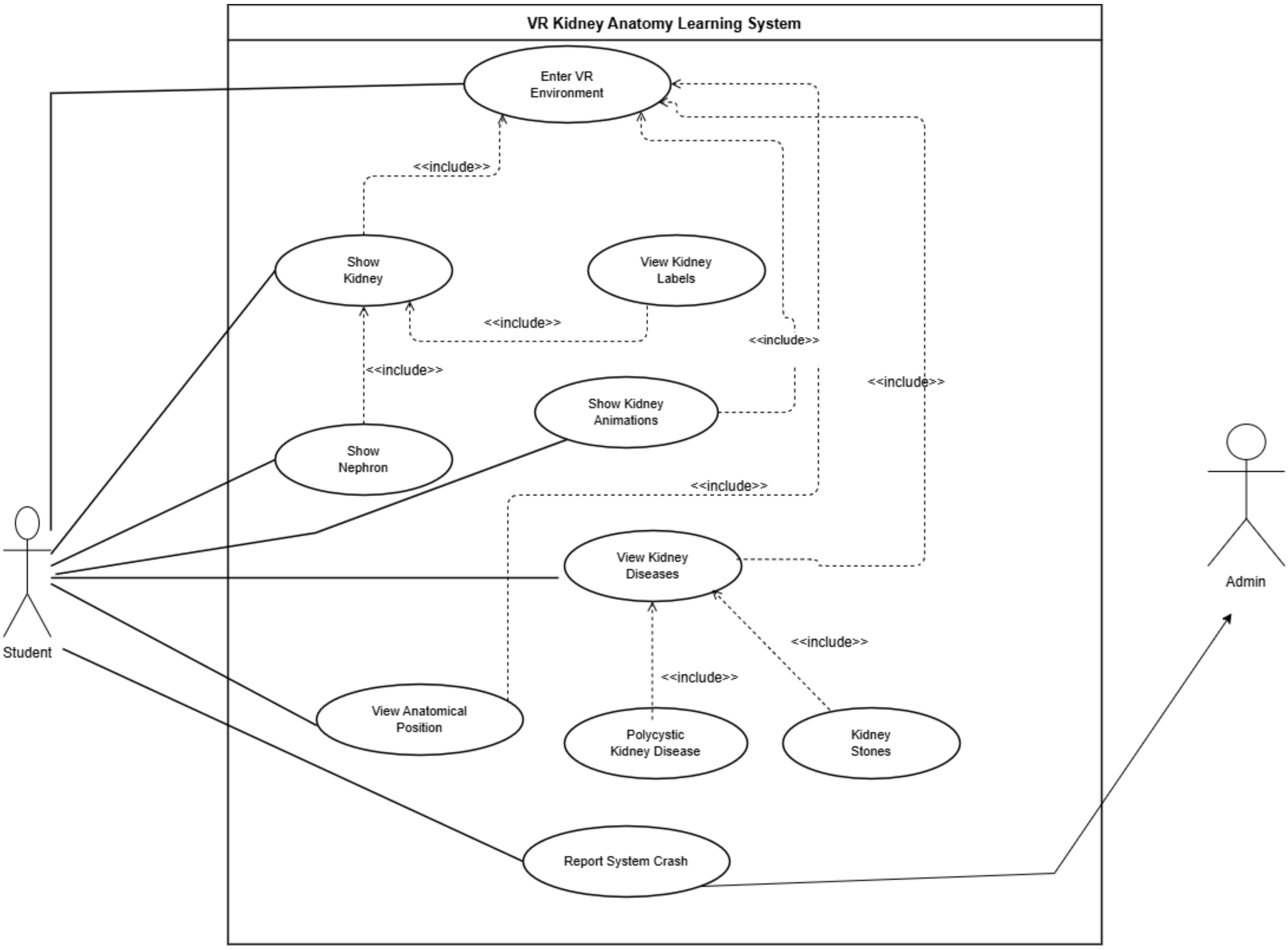

Figure 3.1: Use Case Diagram illustrating the functional requirements of the System

3.7.1 Use Case:Enter VR Environment

The first interaction is illustrated in the use case diagram, called Enter VR Environment between a Student (the actor) and VR Kidney anatomy learning system.It describes one, underlying use case in which the student is the initiator of the virtual reality experience to start their education session.

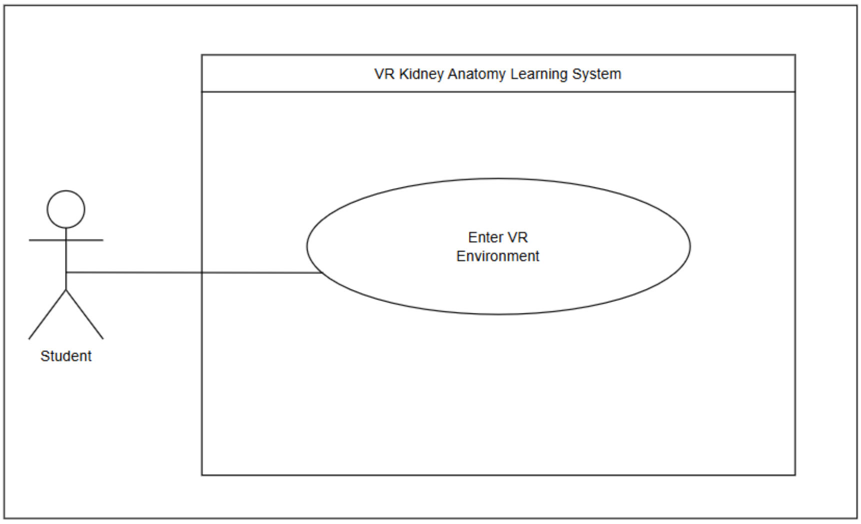

Figure 3.2: Use Case Diagram: Enter VR Environment

Table 3.11: Enter VR Environment Use Case Description

Use Case ID:	UC-01		
Use Case Name:	Enter VR Environment		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	12/01/2026	**Last Revision Date:**	13/01/2026
Actors:	Student		
Description:	User launches the VR application and enters the virtual learning environment.		
Trigger:	Launch VR application		
Preconditions:	VR headset and compatible system must be available.		
Post conditions:	User successfully enters the VR environment and can interact with the learning modules.		
Normal Flow:	**Actor**	**System**	
	1. User wears VR headset.		
	2. User launches the application.	3. System loads the VR environment and user interface.	
		4. System displays the interactive learning menu.	
Alternative Flows:	VR device not detected: System displays an error message.		

3.7.2 Use Case:Show Kidney

This use case diagram illustrates the "Show Kidney" capability within a VR Kidney Anatomy Learning System. It has a Student as the main actor who interact with the system to visualize a 3D model of kidney to be explored educationally. This is activated when the student presses a button "Show Kidney" which has a required dependency (an include relationship) on the "Enter VR Environment". When it is turned on, the system loads and opens an interactive kidney model, provided the user is already within the virtual environment.

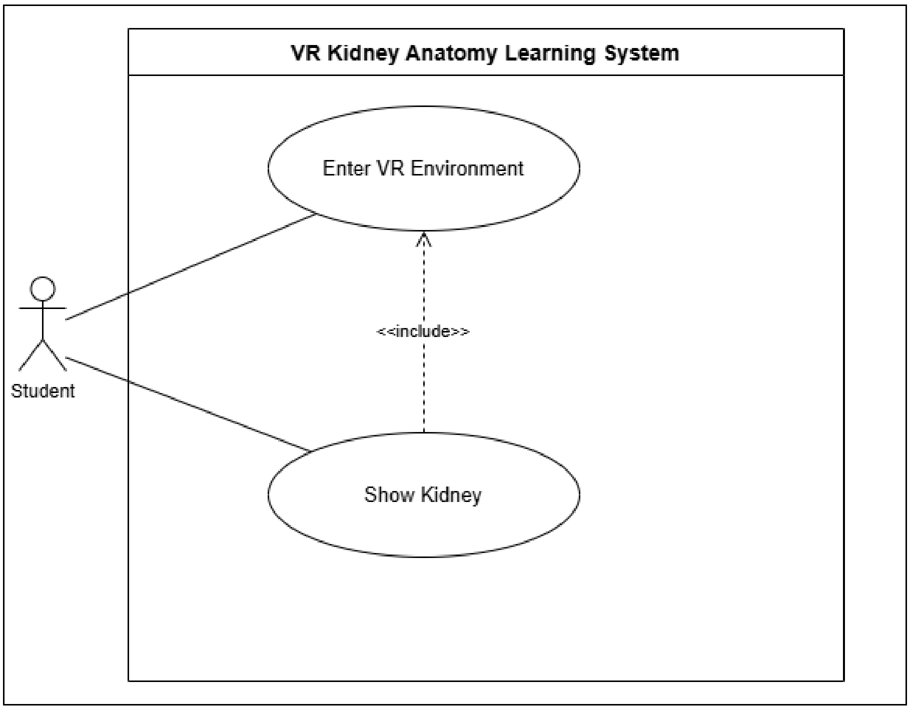

Figure 3.3: Use Case Diagram: Show Kidney

Table 3.12: Show Kidney Use Case Description

Use Case ID:	UC-02		
Use Case Name:	Show Kidney		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	12/01/2026	**Last Revision Date:**	12/01/2026
Actors:	Student		
Description:	User displays the kidney model in the VR environment for detailed exploration.		
Trigger:	Press "Show Kidney" button		
Preconditions:	User must be inside the VR environment.		
Post conditions:	Kidney model becomes visible and interactive.		
Normal Flow:	**Actor**	**System**	
	1. User presses "Show Kidney" button.	2. System loads the kidney 3D model.	
		3. System displays the kidney in front of the user.	
Alternative Flows:	Kidney model fails to load: Nothing gets displayed.		

3.7.3 Use Case:View Kidney Labels

This use case diagram explains the "View Kidney Labels" feature, where a Student can view the Kidney Labels to examine the internal organs of a kidney in VR. When a user grabs a specific kidney component, the system separates the meshes and shows corresponding anatomical name and descriptions. It is also noteworthy that in this use case there is the "Show Kidney" process, meaning that the kidney model has to be in place and in sight to be labeled.

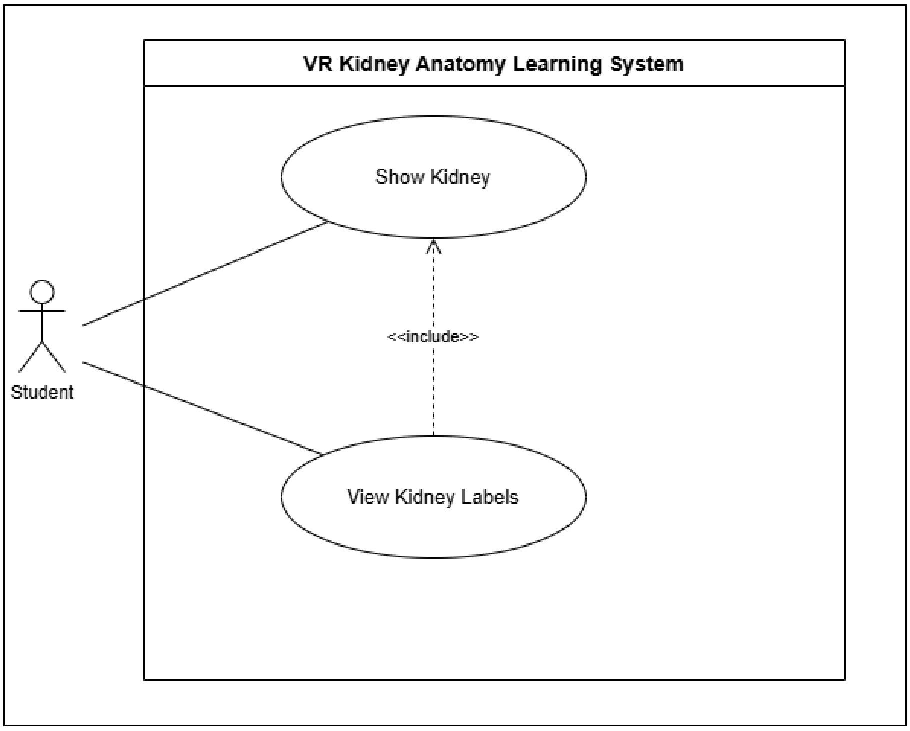

Figure 3.4: Use Case Diagram: View Kidney Labels

Table 3.13: View Kidney Labels Use Case Description

Use Case ID:	UC-03		
Use Case Name:	View Kidney Labels		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	14/01/2026	**Last Revision Date:**	14/01/2026
Actors:	Student		
Description:	User separates the kidney meshes to explore internal structures and view labels and descriptions of each part.		
Trigger:	User selects kidney components		
Preconditions:	Kidney model must be visible.		
Post conditions:	Kidney internal parts are separated and labeled.		
Normal Flow:	**Actor**	**System**	
	1. User grabs a kidney component.	2. System separates internal kidney meshes.	
		3. System displays label and description.	
Alternative Flows:	Mesh separation fails: System reloads kidney model.		

3.7.4 Use Case:Show Nephron

This use case diagram illustrates the "Show Nephron" feature, in which a Student collaborates with the system to research the functional unit of the kidney in VR. The process is initiated by pressing a button,leading the system to load and show a detailed 3D nephron model for interactive exploration. This use case strictly includes the "Show Kidney" functionality, creating the fact that the kidney model must be active as a prerequisite for viewing the microscopic nephron structure.

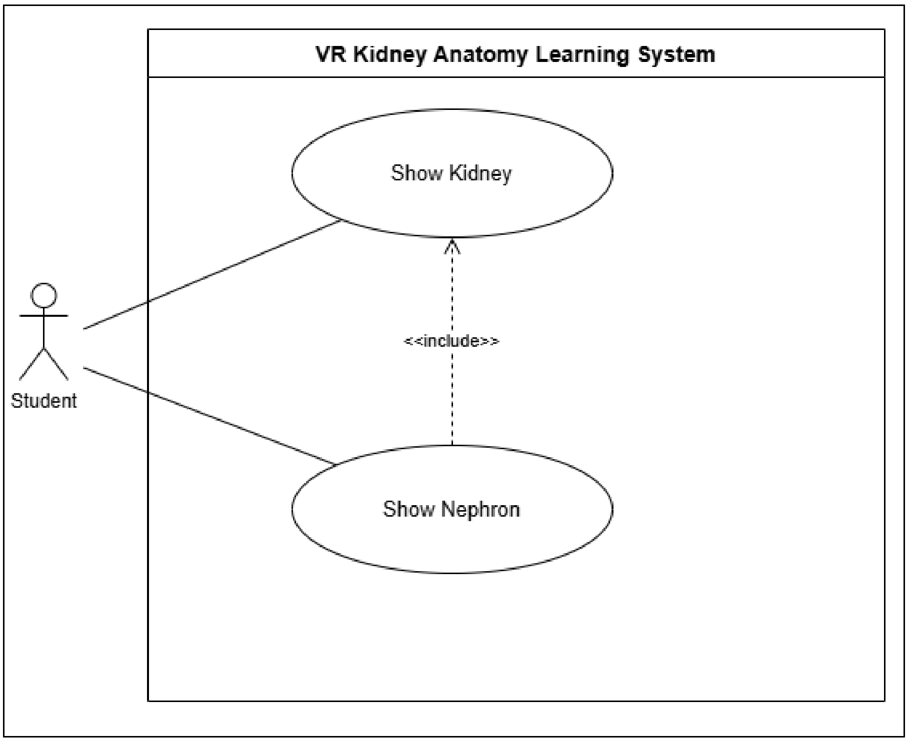

Figure 3.5: Use Case Diagram: Show Nephron

Table 3.14: Show Nephron Use Case Description

Use Case ID:	UC-04		
Use Case Name:	Show Nephron		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	15/01/2026	**Last Revision Date:**	15/01/2026
Actors:	Student		
Description:	User views the nephron model to study the functional unit of the kidney.		
Trigger:	Press "Show Nephron" button		
Preconditions:	For activating the "show nephron" button, the kidney must be explored by clicking the "view kidney" button.		
Post conditions:	Nephron model appears in VR for exploration.		
Normal Flow:	**Actor**	**System**	
	1. User presses Show Nephron button.	2. System loads nephron 3D model.	
		3. System displays nephron for interaction.	
Alternative Flows:	Nephron fails to load: No model is displayed		

3.7.5 Use Case:Anatomical Position

The diagram below is a use case diagram that represents the "Anatomical Position" feature, which enables a Student to see the exact position of kidney in a model of a complete body of a human being. When Upon a button press, the system will cause the camera to move within the virtual body to highlight the location of the kidney. This process includes "Enter VR Environment" use case, which requires one to be within the virtual space for viewing the anatomical model.

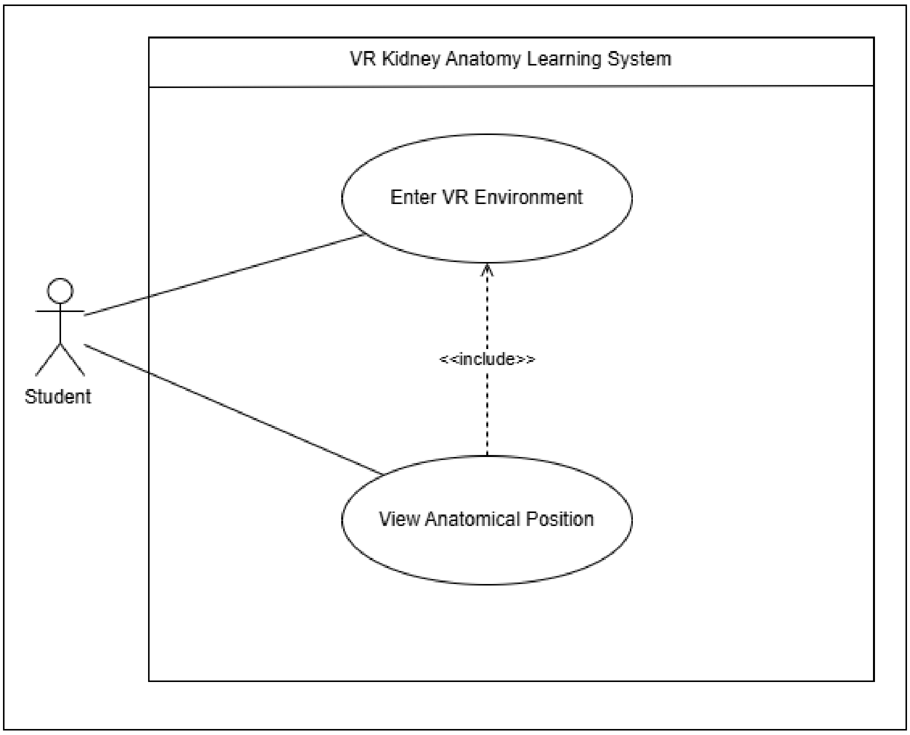

Figure 3.6: Use Case Diagram: Anatomical Position

Table 3.15: Anatomical Position Use Case Description

Use Case ID:	UC-05		
Use Case Name:	View Anatomical Position		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	21/01/2026	**Last Revision Date:**	21/01/2026
Actors:	Student		
Description:	User views the anatomical location of the kidney within the human body.		
Trigger:	Press "Anatomical Position" button		
Preconditions:	User must already be inside the VR environment.		
Post conditions:	User views the kidney's position inside the human body.		
Normal Flow:	**Actor**	**System**	
	1. User presses Anatomical Position button.	2. System displays the full human body model.	
		3. System moves camera inside.	
		4. System highlights kidney.	
Alternative Flows:	Body model fails: error message shown.		

3.7.6 Use Case:Kidney Function Animation

The "Kidney Function Animation" feature is represented in this use case diagram where a student studies how kidney works.Upon pressing the "Show Animation" button, the system plays a 3D animation with an audio to clarify and to improve the learning process. This functionality includes the "Enter VR Environment" use case.The student is required to be active in the virtual space,for the animation to load and display.

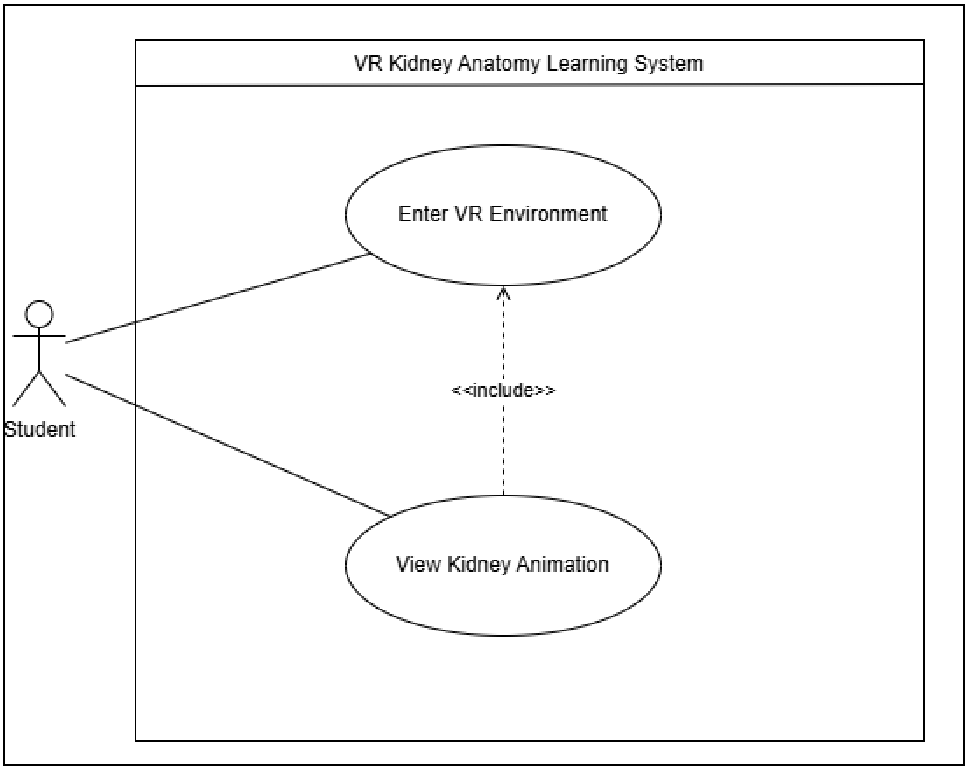

Figure 3.7: Use Case Diagram: Kidney Function Animation

Table 3.16: Kidney Function Animation Use Case Description

Use Case ID:	UC-06		
Use Case Name:	View Kidney Animation		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	21/01/2026	**Last Revision Date:**	21/01/2026
Actors:	Student		
Description:	The system plays an animation demonstrating the filtration process of the kidney along with an audio explanation.		
Trigger:	Press "Show Animation" button		
Preconditions:	User must be inside the VR environment.		
Post conditions:	Animation completes and the user returns to the room view with the navigation bar.		
Normal Flow:	**Actor**	**System**	
	1. User presses Show Animation button.	2. System starts the kidney function animation.	
		3. System plays audio narration.	
		4. Animation ends.	
Alternative Flows:	Animation fails to load: System displays an error message.		

3.7.7 Use Case:Kidney Disease Visualization

This use case diagram shows the feature of Kidney Disease Visualization which allows a Student to explore different diseased kidney models in VR. Through the choice of certain options such as "Polycystic Kidney Disease" or "Kidney Stones" the system loads the relevant 3D model and with an informative audio description.

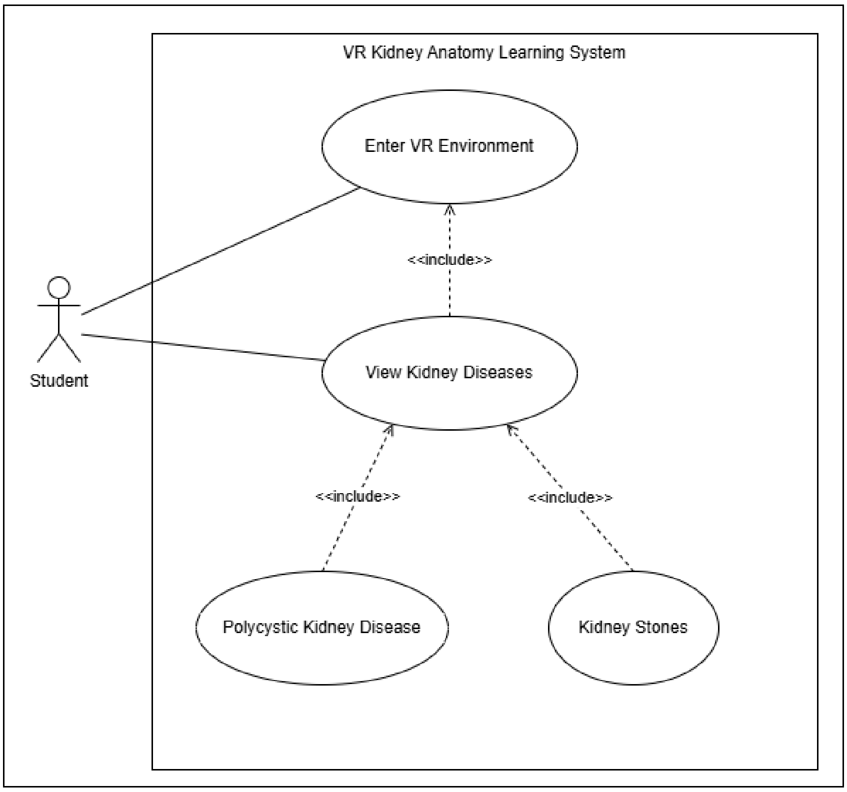

Figure 3.8: Use Case Diagram:Kidney Disease Visualization

Table 3.17: Kidney Disease Visualization Use Case Description

Use Case ID:	UC-07		
Use Case Name:	View Kidney Diseases		
Created By:	Areeba Amin	**Last Updated By:**	Atika Abid
Date Created:	21/01/2026	**Last Revision Date:**	21/01/2026
Actors:	Student		
Description:	User explores kidney diseases by selecting different diseased kidney models.		
Trigger:	Press "Show Diseases" button		
Preconditions:	User must be inside VR environment.		
Post conditions:	Selected disease model is displayed with audio explanation.		
Normal Flow:	**Actor**	**System**	
	1. User presses Show Diseases button.	2. System displays disease options.	
	3. User selects Polycystic Kidney or Kidney Stones.	4. System loads the diseased kidney model.	
		5. System plays audio explanation.	
Alternative Flows:	Disease model fails to load.No model is displaed		

3.8 Project Feasibility

The proposed VR-based kidney anatomy learning system is feasible in terms of technical, operational, economical and time aspects.
Technical Feasibility: The system can be developed using Unity 3D as the primary VR development platform and Blender for 3D modeling and optimization. The application is compatible with a standard VR headset, including Oculus or Vive, and a compatible computer system. Since the development team already has experience working with Unity and blender.The technical specifications of the project can be efficiently managed.
Operational Feasibility: The system is designed to be practical and user friendly for medical students. Users can interact with the virtual environment using intuitive controls, which will enable them to study the structures of kidneys, see the labeled parts and observe functional animations. This interactive method makes learning an interactive process and more interesting and comprehensible than the traditional static learning techniques.
Economic Feasibility: VR headsets and certain 3D models might require preliminary costs, many development tools used in the project, such as Unity and Blender, are available for free. Moreover, there are a number of 3D assets, which can be obtained at no or low costs.The virtual simulation can be reused severally without extra cost once developed,making it cost effective educational tool in the long run as compared to traditional cadaver-based learning.
Schedule Feasibility: The project was broken down into several development phases to ensure timely completion. These steps involved developing models, kidney exploration, interactive animations, nephron exploration.In addition disease simulations like polycystic kidney disease and kidney stones formation were introduced to improve educational value. A sufficient time was set aside to develop, test, and improve, making sure that the project must be completed within the given timeframe.

3.9 Summary

This chapter discussed the shortcomings of the current anatomy learning systems,which emphasized the focus on static visualizations, lack of functional understanding, and the complexity of the mechanisms of interaction. In order to overcome these shortcomings,a VR-based Anatomy Learning System focused on kidney visualization was proposed which offers a 3D navigation, 3D mesh separation, descriptive labeling,and physiological process animation.The chapter provided the non-functional and functional requirements as well as targeted users and features that were intended to enhance usability, engagement,and educational value. The system architecture, workflow diagram will be displayed in the following chapter and detailed design to aid in implementation.

CHAPTER 4

DESIGN

The VR Kidney Design System will focus on converting the functionality requirements into a collection of virtual learning that is scalable, systematic and interactive. This chapter describes the design constraints, system architecture, design methodology that have occurred to develop an efficient, user-friendly and reliable application. Both the high level and the low level design factors have been discussed which are used to describe the elements of the system, their interaction and technical background of the system. The design is easy to use and smoothly 3D-rendered, anatomically accurate, and immersive user experience along with being modular, scalable, and VR-compatible hardware.

4.1 System Architecture

Figure 4.1 illustrates the general system structure of VR Anatomy Learning project. The system has a structured process which begins as shown in the diagram and wraps up with the last deployment on the Meta Quest 2 headset.
The design is structured as a flow of work to make sure that the raw 3D models are effectively incorporated with the interactive system logic. The first step is the process of the Data Providers stage, in which quality anatomical models and medical references are found sourced. This is transferred to the Data Gathering phase where all the required 3D resources are prepared and gathered towards the project.
Then, the system goes into the Asset Preparation and Pipeline step. This is an important aspect of the project in which static models are cleaned, optimized and exported out of Blender into Unity. At this phase, complicated anatomical meshworks are refined to make them simple enough to be efficiently used in a virtual reality world without interrupting the performance.
After the preparation of the assets, they are incorporated with the system logic. In this phase,the 3D models are made interactive learning objects by linking them with static objects. custom C# scripts. The system can identify user actions by these scripts, including pointing at objects either through raycasting or by taking objects physically in the VR world. Figure 4.1 also demonstrates the way various internal elements, e.g. State Management, collaborate in order to deliver a seamless experience. In this module, the control is made on which anatomy structures are now in operation so that only a part of it like the Kidney is taken or Nephron is presented to uphold good performance.
Lastly, the stage of User Interface and Experience unites all. The Unity engine renders all interactions and visualizes in a stereoscopic (3D) virtual reality view. This organized methodology enables the system to be scalable and provides quality performance with the few resources of a mobile VR device.

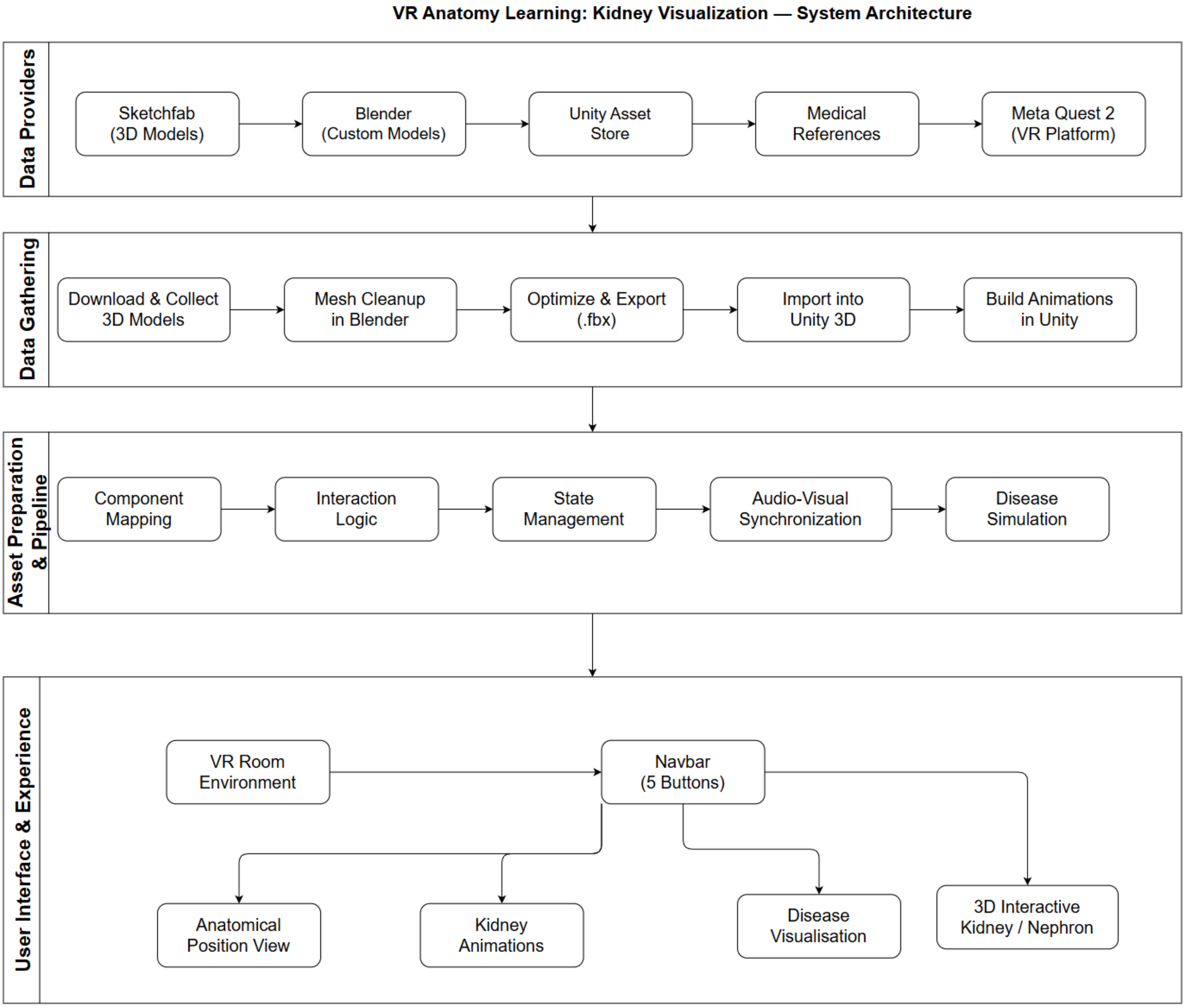

Figure 4.1: System Architecture of the VR Kidney Visualization System.

4.2 Design Constraints

Several technical aspects were considered when designing the VR Anatomy Learning system and practical constraints to ensure a consistent user experience:

1. **Hardware Constraints:** Since Meta Quest 2 is an independent mobile VR headset, the 3D models were optimized in Blender to ensure low draw calls and have a low maintenance consistent 72 FPS.
2. **Spatial UI and Visual Clarity:** VR users are able to look around. We had to make sure that the text panels were big and could be read but placed in a way that they did not intersect or mask out the parts of the kidney.
3. **Performance Constraints:** The size of textures and real-time lighting were restricted to remain in the thermal and memory limits of the device.
4. **Human machine interaction:**A camera system that prevents exploring motion sickness should be implemented.

4.3 Design Methodology

VR Kidney Visualization system was developed using the Incremental process. Development Model as shown in Figure 4.2. The choice of this model was to enable the five modular stages of which the project could be constructed to ensure that every feature was completely tested and working in one stage before proceeding to the next.

4.4 High Level Diagram

The High-Level Diagram (HLD) is an overview of the system operation flow, which shows the interaction of the core modules in the virtual environment. It focuses on the change over the system start-up and hardware, monitoring the processing of user interface with the VR Navbar. The HLD maps out these main processing branches, by mapping them out enables the application to be very modular allowing the application to transition smoothly between interactive animations and simulation of diseases, anatomical exploration. This stepwise procedure of application deployment to the ultimate Meta rendering is detailed in this article.The High Level Diagram is illustrated in Figure 4.3

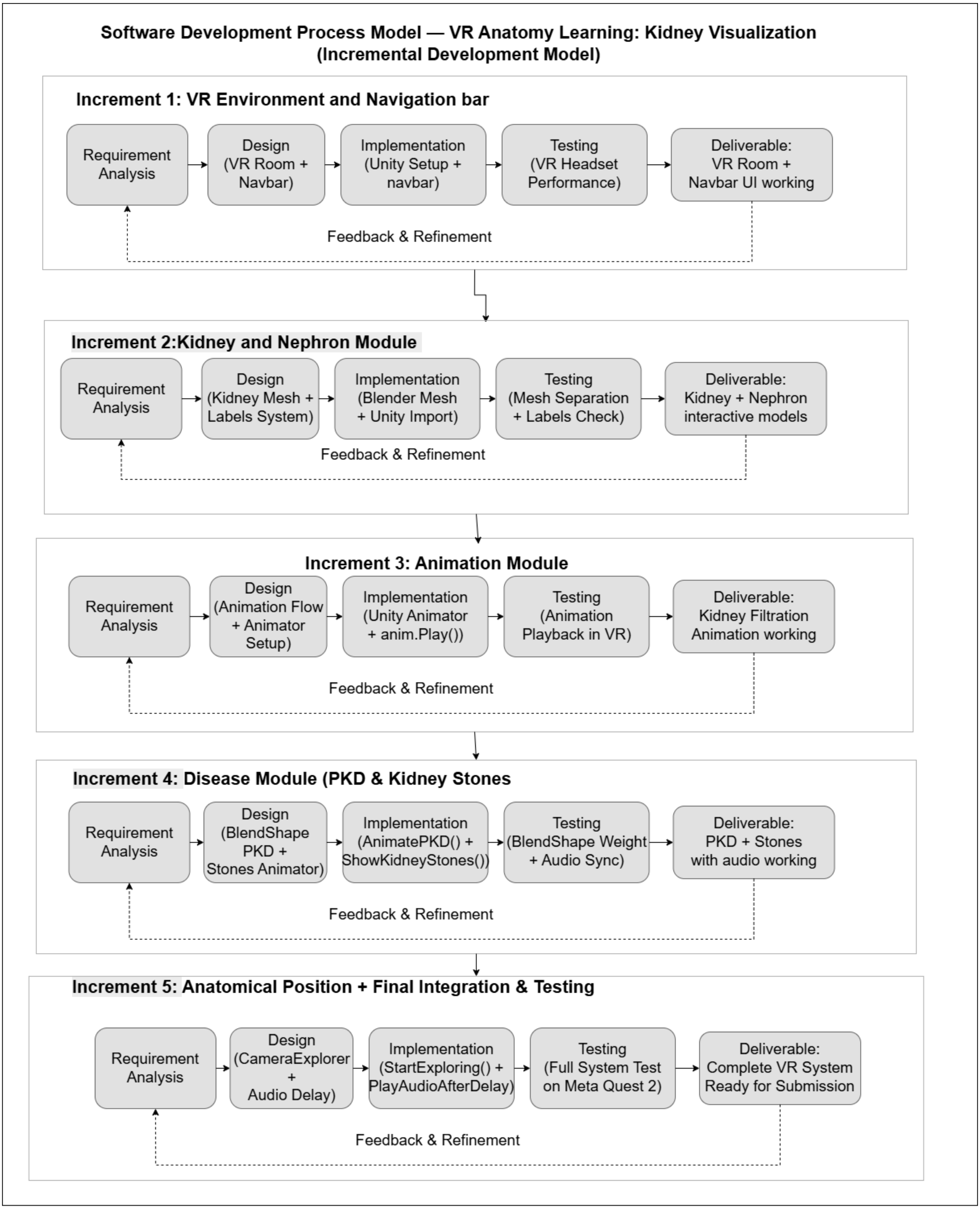

Figure 4.2: The Incremental Development Model used for the VR Anatomy Learning system.

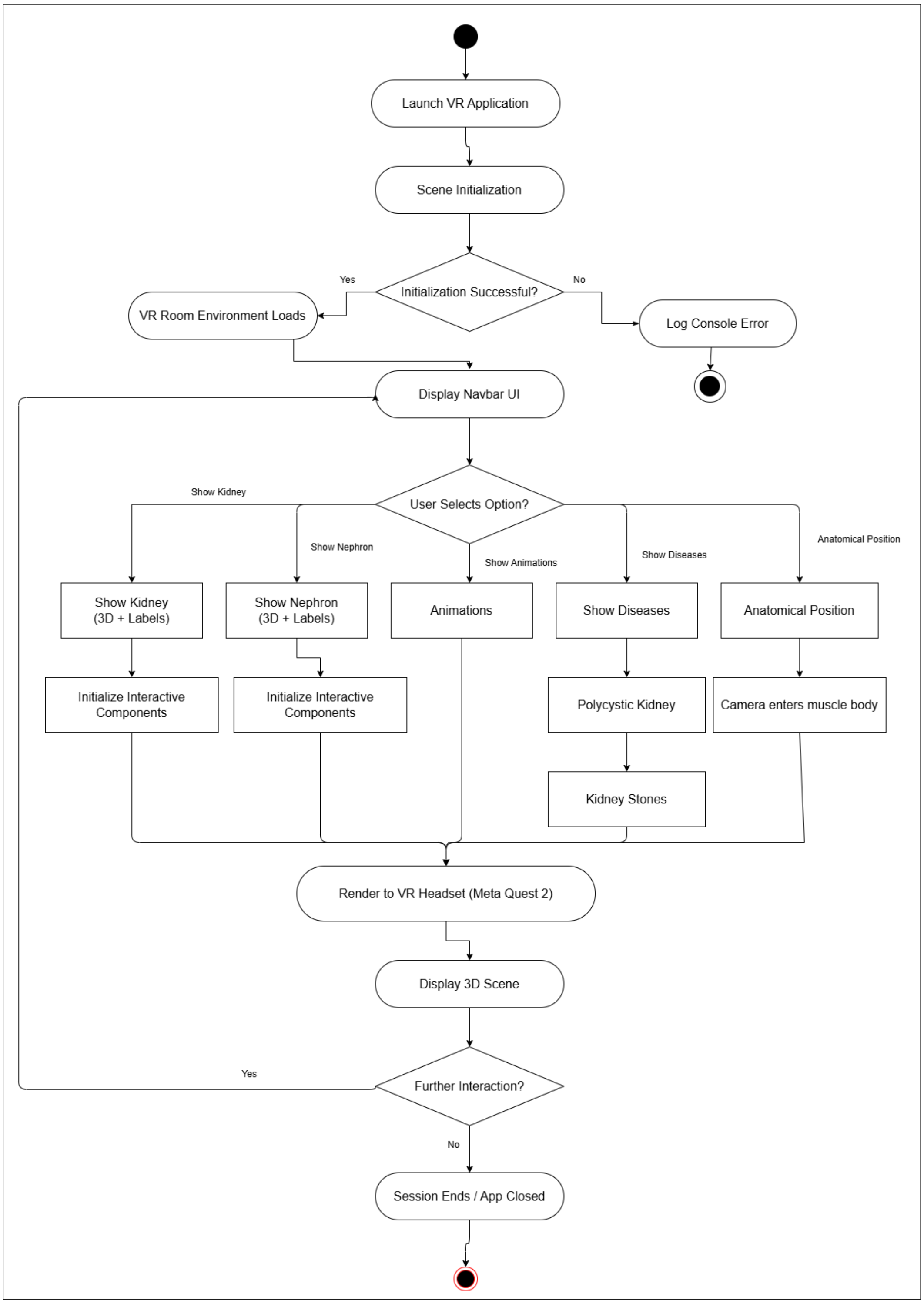

Figure 4.3: High-Level Diagram (HLD)

4.5 Low Level Diagram

The low-level Diagram determines the finer structure of every component of the system by breaking them down the system to smaller modules and functional units. The system is made by these modules less difficult to develop, maintain and test. The modules in this project deal with interactions among the 3D anatomical models, user interaction scripts, and XR interaction scripts. This is a modular design that enables easy interaction with the user, real time updates and efficiency in implementation of animations and simulations of diseases. The below figures depict the activity flows of the main system functions.In Figure 4.4, Shows the steps of entering the VR environment.

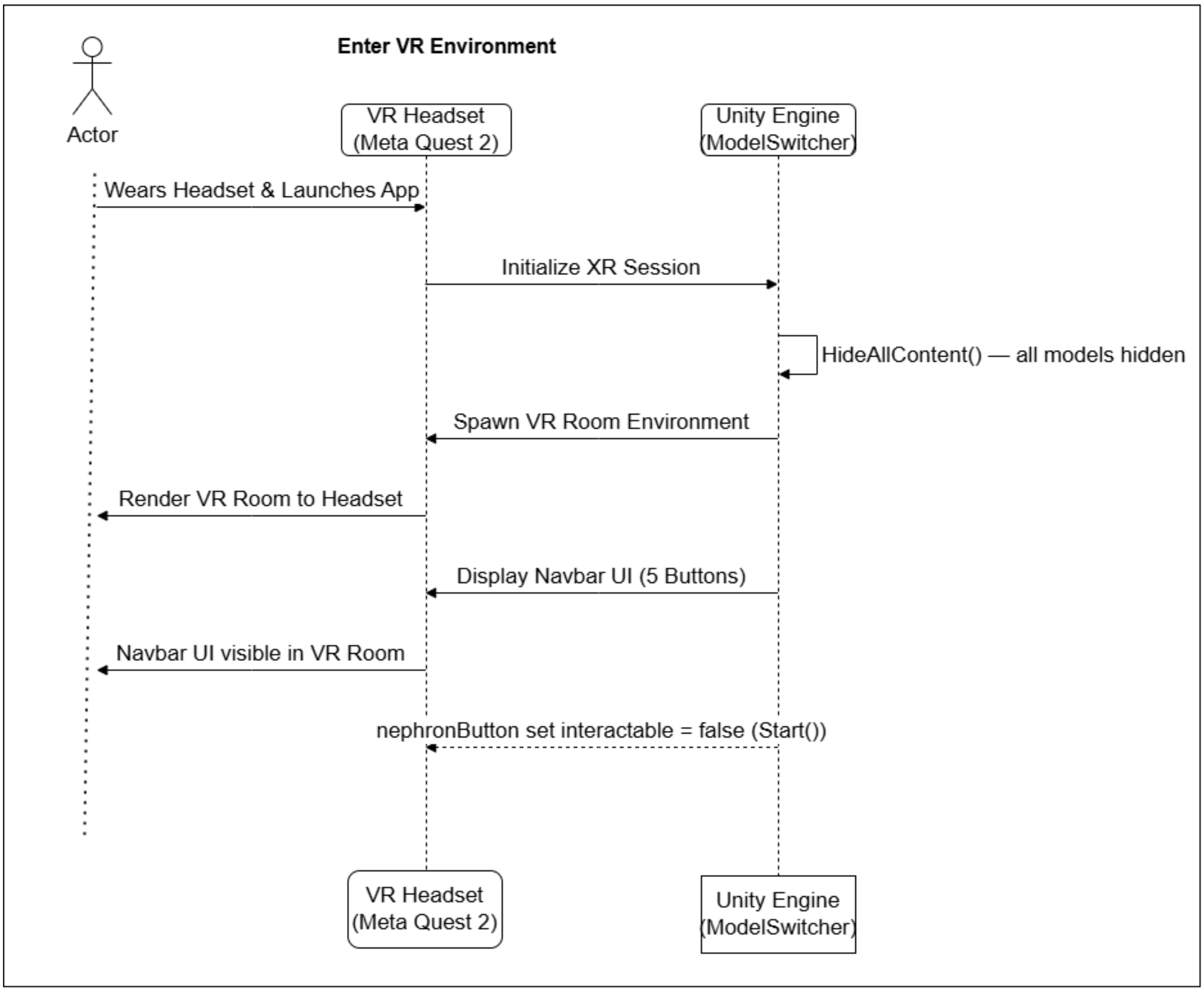

Figure 4.4: Low Level Diagram for Entering the VR Environment

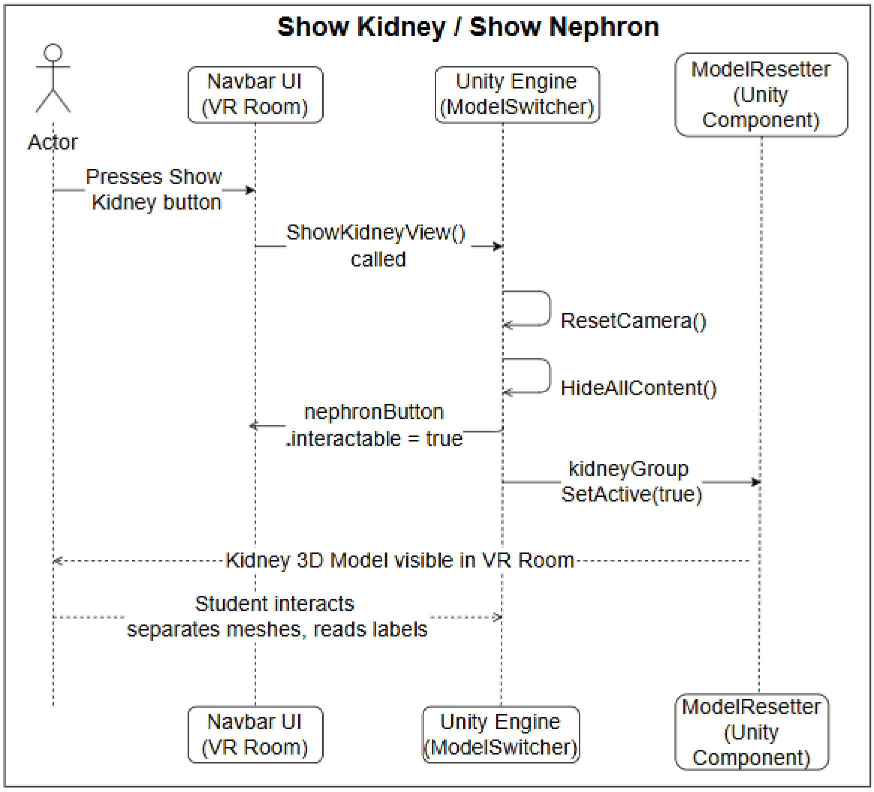

Figure 4.5: Low Level Diagram for Showing Kidney and Nephron

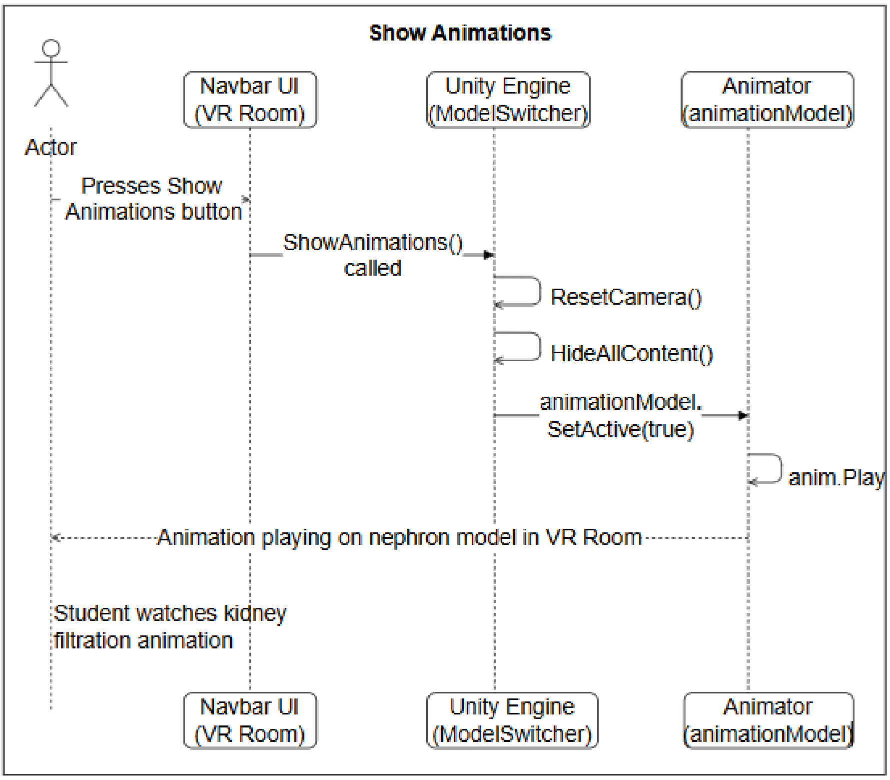

Figure 4.6: Low Level Diagram for Showing Animations

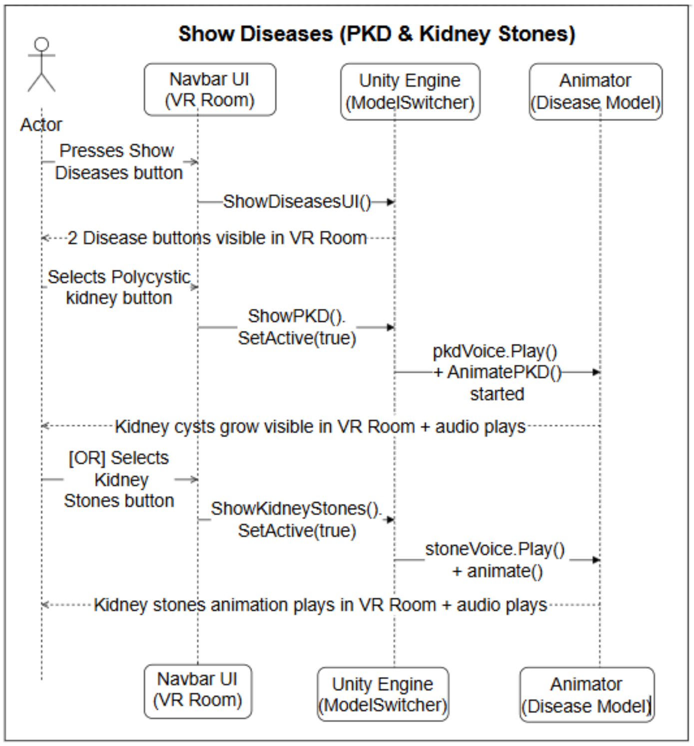

Figure 4.7: Low Level Diagram for Showing Diseases

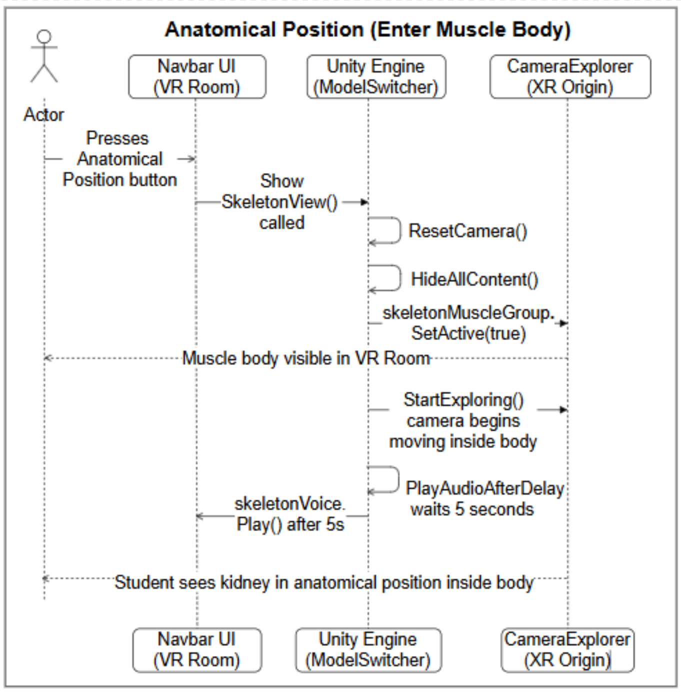

Figure 4.8: Low Level Diagram for Anatomical Position Visualization

4.6 Data and Asset Management

In contrast to traditional applications which rely on external servers, this application uses the local Unity Asset Database as opposed to relational SQL database. The choice of this design is to satisfy the performance needs of the Meta Quest 2 headset. Using 3D models, textures, and by saving them. The system can access audio files stored locally within the application, and does not require any time network delays. This can facilitate an enjoyable and interactive VR learning experience.
The system handles anatomical data with Serialized C# data structures as well as Unity built-in resource management. The system is linked to interactive scripts and their corresponding 3D models with the identifiers of the objects instead of utilizing database queries. This enables the system to rapidly alternate the various states, including a normal kidney and diseased kidney such as Polycystic Kidney Disease (PKD).
Moreover, file formats like FBX (3D models), MP3 (audio), are optimized so that can be used to enhance performance. The application can be used in a way that is completely offline since everything is stored in the area, thus it can be used in medical education and demonstrations even when there is no access to internet.

4.7 Class Diagram

Fig. 4.9, Class Diagram gives an overall picture of the target system by explaining the objects and classes in the environment and the relationships between them. The possibilities of using this model are very wide from first conceptualising the specific data structures to advanced design of the software architecture. By defining the properties and algorithms of the main C# scripts, it assures a flexible and modular VR anatomical simulations.

4.8 GUI and Interaction Design

The Graphical User Interface (GUI) is a implemented Spatial UI, which has menus to live in the natural state in 3D virtual world. This design has the focus on immersion and ease of use, making sure that learning does not hinder the educational content of user's field of view. The interface mainly consists of a floating navigation bar and context aware identifications of certain anatomical exploration.
The figures below give a visual impression of the interface of the system and show the change between the original setting of the environment to the detailed anatomical and pathological visualizations.

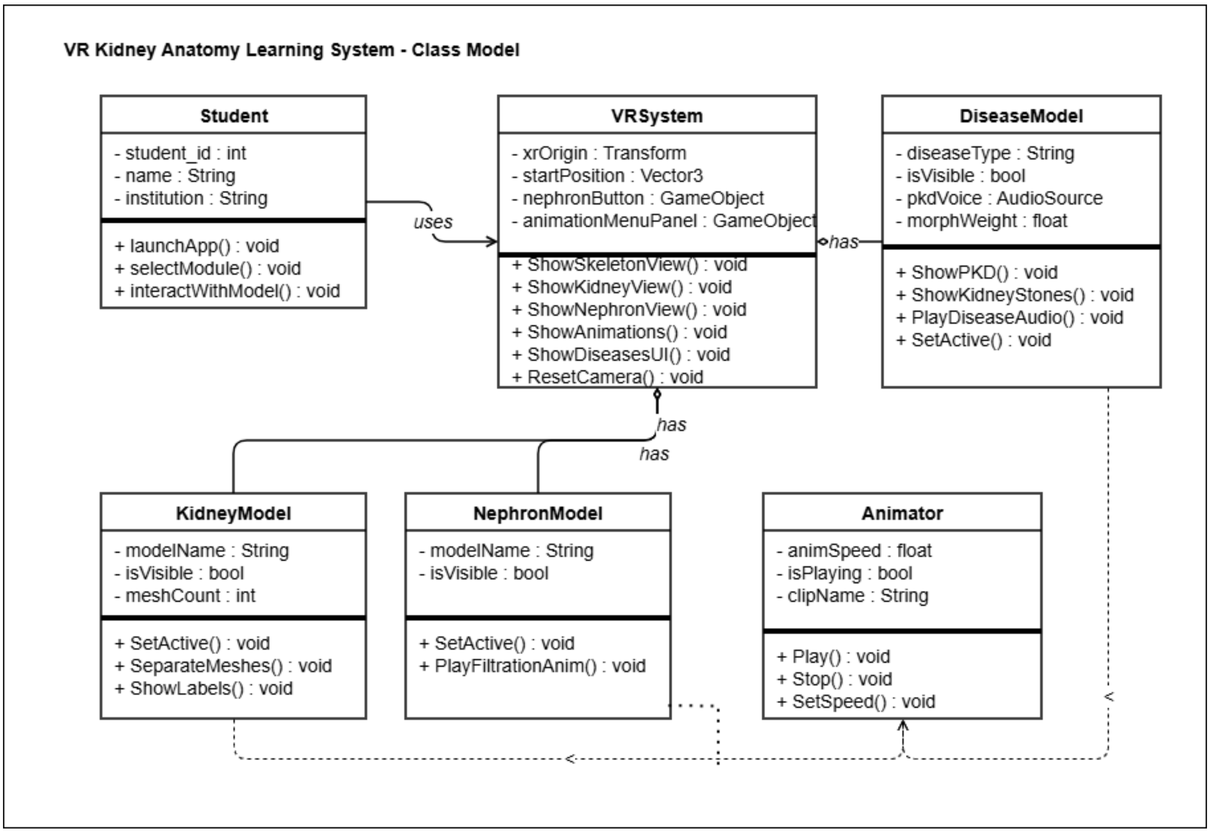

Figure 4.9: Class Diagram

Figure 4.10: UI Environment featuring the Central Navigation Hub.

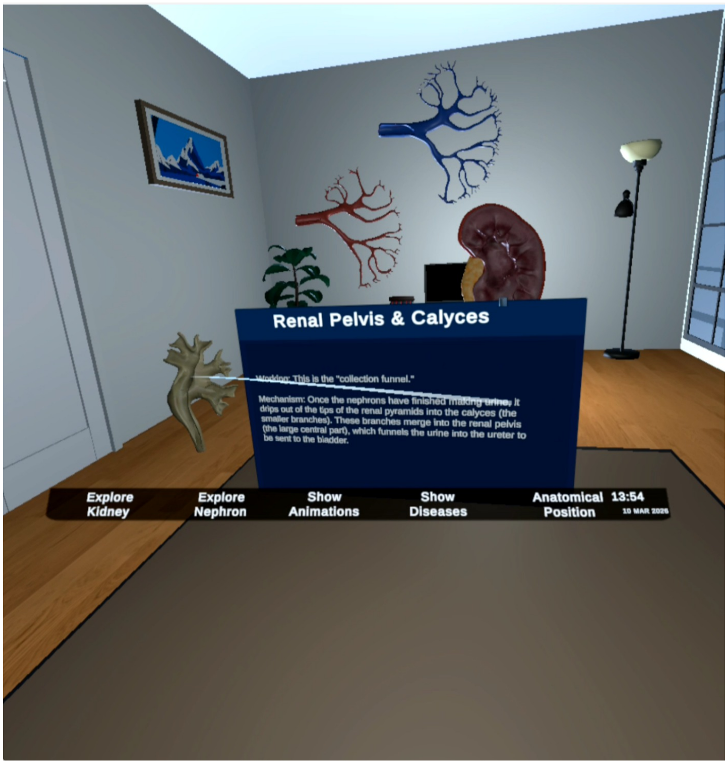

Figure 4.11: Interactive Kidney Module demonstrating Mesh Manipulation and Labelling

Figure 4.12: Microscopic Visualization of the Nephron Structure within the VR Space.

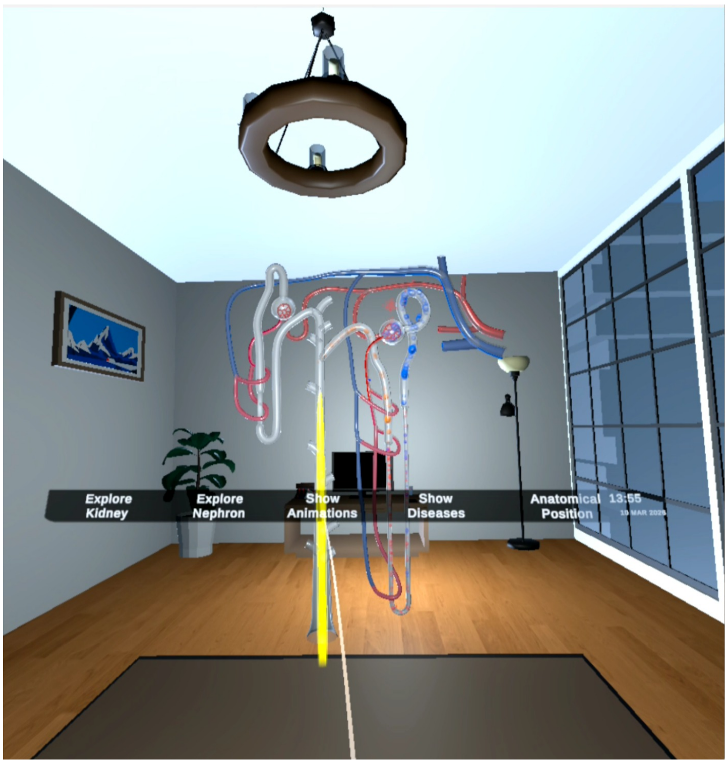

Figure 4.13: Animation sequence illustrating the Anatomical Process Flow.

Figure 4.14: Interface with specialized buttons for Disease Simulation.

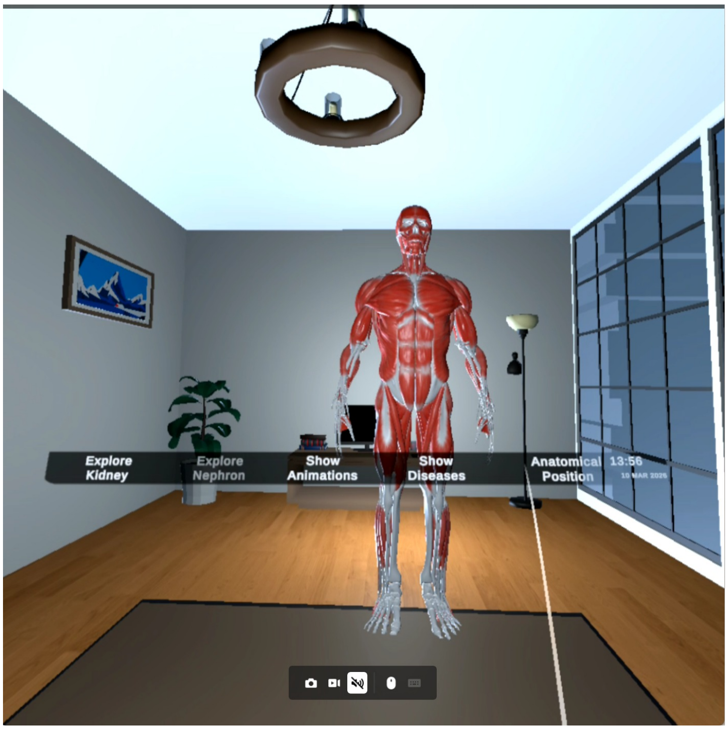

Figure 4.15: Anatomical Position Perspective showing the Muscular Body.

4.9 Summary

In this chapter, the general design of the VR Kidney Visualization system was given. It explained the system architecture and how different components work together to create a virtual learning environment that is interactive. The key design has also been discussed in the chapter.Limitations of VR hardware functionality, spatial user interface design, and user comfort.

Development was based on the Incremental Development Model in order to develop the step by step with testing of each feature. High level and low level designs were detailed to give an understanding of the system workflow and how various modules interact with each other as environment entry, model switching, animations, and disease visualization. Lastly, the chapter described the data and asset management strategy and the GUI design used in the virtual environment. These elements in combination in design make sure that the application is efficient, scalable and can supply an immersive learning experience on the Meta Quest 2 VR platform.

CHAPTER 5

SYSTEM IMPLEMENTATION

The implementation stage is where the conceptual design of the proposed system is implemented and converted into a working application. During this stage, the architecture design made in the earlier chapters is developed with the help of suitable software tools and technologies.The implementation process integrates 3D models of anatomies in order to produce an interactive virtual environment, and allowing user-interaction with it.
In the project VR Anatomy Learning: Kidney Visualization, the implementation is concentrated on creating an immersive learning experience in which students will be able to study the human kidney and its structures in three dimensions. The system is created with the help of the Unity game engine along with the XR Interaction Toolkit to support VR-based interaction. It allows users to navigate through various learning modules, and interacts with anatomical structures and watch simulations of kidney functions and diseases.

5.1 Development Tools and Technologies

A number of the tools and technologies are involved in the creation of the VR anatomy learning system:Kidney Visulaization.

1. **Unity Game Engine:** Unity is the main development platform used to create the virtual environemnt. It supports real time rendering, physics simulation,VR integration, and animation control.
2. **C# Programming Language:** C# language is used to implement the functionality of the application. It manages object associations, user interface, animation triggers, and other system processes.
3. **XR Interaction Toolkit:** XR Interaction Toolkit is the Unity framework that is used to enable VR interactions in the application. It allows the implementation of features including grabbing,selecting and interacting with the UI inside virtual reality environment.
4. **Blender:** Blender is used to pre-process and optimize the 3D anatomic models before introducing them to Unity. Mesh cleanup, adjusting of materials and model optimization are done in Blender.
5. **Meta Quest Headset:** Meta Quest is a series of wireless, all-in-one virtual reality (VR) and mixed reality (MR) headsets created by Meta Platforms. The virtual reality application is developed on the Meta Quest headset. This gadget enables learners to experience the virtual learning environment through immersive 3D visualization and motion interaction.

5.2 System Modules

The system is subdivided into a number of modules which offer various learning modes.

5.2.1 Navigation Module

Navigation module gives the primary interface where the user can access various features of the system. In the virtual environment, there is a navigation bar with several buttons which enable the user to switch between various learning modes. These buttons include:

- Show Kidney
- Show Nephron
- Show Diseases
- Anatomical Position
- Kidney Animations

Every button runs a particular module or visualization in the VR environment.

5.2.2 Kidney Visualization Module

The kidney visualization module enables users to learn about detailed 3D model of the human kidney. The model is broken down into a series of meshes which represent various anatomical structures.

Users can interact with these meshes using VR controllers. When a structure is grabbed, a description panel will appear describing the purpose of that part. This interactive procedure assists the students to comprehend how the kidney is internally structured.

5.2.3 Nephron Exploration Module

The nephron module gives an even more in-depth look at the functional unit of the kidney. An individual 3D model of the nephron is shown where users can view components such as the glomerulus, tubules and collecting ducts.The user can customarily interact with various areas of the nephron.This will enable the students to know what filtration is and the role of each part in the formation of urine.

5.2.4 Disease Visualization Module

The system has an illustrative module that shows two diseases that affect the kidneys:

- Polycystic Kidney Disease (PKD)
- Kidney Stones

The simulation of kidney stones has small particles that merge into a large stone. This animation shows the visual explanation of the formation of mineral deposits in the kidney in a more detailed manner.

The PKD simulation demonstrates the development of cysts and their impact on normal structure of the kidney.

5.2.5 Kidney Function Animation Module

This module demonstrates the functional behavior of the kidney through animated sequences. These animations illustrate processes of urine formation.

Watching these dynamic visualizations, students will be able to comprehend the workings of the kidney better and understand the physiological functions.

5.3 User Interaction Implementation

The interaction with the user is applied with the help of the XR Interaction Toolkit in Unity. Each anatomical structure is defined with interaction components enabling the system to detect user actions.
The main interaction techniques include:

1. **Grab Interaction:** Users can grab anatomical structures using VR controllers to observe them more closely.
2. **Selection Interaction:** Selection is used to trigger the buttons in navigation bar and to switch between different modules.User can press buttons such as show kidney,show nephron,Anatomical Position,Show diseases,Show Animation.

These interactions provide a natural and intuitive learning experience.

5.4 Asset Management

Any visual files in the system are stored in the Unity asset management system. These assets consist of 3D models, textures, scripts, animations and audio files. The anatomical models are imported as FBX files and converted into prefabs for easy reuse within the virtual environment. Materials and textures are added to enhance visual realism. There are instances where transparency is used on structures like Bowmans Cpasule to show internal structures, such as the Glomerulus.

5.5 System Workflow

The VR anatomy learning system has a structured workflow.

1. The application starts by loading the virtual environment.
2. The user uses the navigation panel to choose a learning module.
3. Depending on the chosen option, the related 3D model or animation is loaded into the scene.
4. The user interacts with anatomical structures using VR controllers.
5. Labeling, information panels, and animations are activated according to the user actions.

5.6 Summary

The implementation phase of the system is aimed at converting the proposed design into a working VR application. Combining Unity with XR Interaction Toolkit and optimized 3D anatomical models,the system offers a hands-on learning experience of kidney anatomy. The modular nature enables users to experiment with the kidney, learn about its internal structures, watch simulations of diseases, and study its physiological functions through interactive visualization. The implemented system is able to prove successful how virtual reality can benefit anatomy education with a more engaging and realistic learning experience.

CHAPTER 6

SYSTEM TESTING AND EVALUATION

6.1 Importance of System Testing

One of the major steps in the creation of the VR Anatomy Learning is system testing: System testing of Kidney Visualization system ensures that everything in the program functions well,intelligibly, and consistently in a virtual setting. The system is based on immersive virtual reality: testing is far more important to enforcing not only functional correctness but also user interaction, performance stability, and responsiveness in real-time. The goal of testing in this project is to make sure that all modules including kidney visualization, nephron exploration, disease simulation, and animation are tested to ensure they works well and provides correct instructional information. In addition, testing can assist in the detection of technical problems such as as lag, bad interaction, or rendering bugs that could worsen the learning experience. System is improved by complete testing and evaluation to make it usable, correct and effective in developing an interactive learning platform.

6.2 Graphical User Interface (GUI) Testing

Graphical User Interface testing is the assessment of the visual and interactive aspects of the virtual reality environment to ensure that users can freely move around and interact with it. The GUI testing in this project involves making sure that the navigation menu, buttons, labels,and virtual panels in the virtual world are functional. Every button, e.g. "Show Kidney, Show Nephron, Show Diseases, Anatomical Position and Kidney Animations," is verified to ensure that it switches on the right module immediately without any errors. Moreover, the layout and location of UI elements are evaluated to guarantee that they can be readily observed and properly positioned in the VR space. Special interaction feedback, including selection responses, is paid attention to make sure that users get distinct visual clues in the process of interaction. This testing will ensure that the interface is friendly, interactive, and suitable to immersive learning.

6.3 Usability Testing

The usability testing is done to find out the ease with which the VR system can be used and the user can reach his or her learning objectives. Since the system is supposed to be used by students with little or no knowledge of virtual reality, it is essential that the system must be user-friendly and require no considerable technical expertise. Usability testing involves observing the behavior of users including entering the VR environment, choosing various modules, interacting with kidney models, and observing animation.

Response to the ease of navigation, instructions clarity, and overall user experience is obtained. The analysis also considers the comfort of motion, ease of interaction, and efficiency in learning. Depending on the feedback, adjustments are done to simplify controls, enhance instructions and streamline interaction flow so that the system provides a smooth and interactive learning process.

6.4 Software Performance Testing

Performance testing identifies the performance of a system in different conditions, particularly in a resource-intensive VR environment. This includes the frame rates assessment, loading times, and interface response to ensure a smooth, lag-free performance. In this project, performance testing is achieved by executing the program on the target VR monitoring system and hardware during different activities like 3D model loading, switching models, and executing animation. High-performance is required to guarantee good performance. Particle systems and polygon models used in the process of renal functions animations are optimized. Specifically, memory usage and processing speed are also tested in the system to ensure that it does not crash or freeze when using it. Such tests help to verify so that users can experience a stable and interactive experience.

6.5 Compatibility Testing

The compatibility testing ensures that the VR application is correctly supported on the suitable hardware and software platforms. Since the system is implemented in Unity and deployed on a VR headset, it is essential to make sure that every functionality works correctly on the target device. It is tested to make sure that the application is functioning correctly on the Meta Quest VR headset and compatible with the XR Interaction Toolkit. The system is also verified to work with many configurations, including hardware differences in performance and display options. This will ensure that the functionality of the application and visual quality do not change, even with minor system configuration. The compatibility testing can be used to assure reliability and accessibility of the system to all intended users.

6.6 Exception Handling

Exception handling testing is done to make sure that the system is able to deal with unexpected situations or user errors without crashing or giving incorrect responses. This is an experiment on a variety of situations, such as wrong interaction, rapid change of models, etc. freezes in the process of animation. The system will be able to cope with such incidents in a graceful way by providing relevant feedback to the user. For example, when a user attempts to interact with an object that is not interactive, the system ensures that there are no surprises that occur. Likewise precaution is observed to avoid overlapping computer-generated images and improper loading of models. This testing ensures that the application is strong, consistent and effective in a diverse range of situations.

6.7 Test Cases

Test cases are supposed to systematically validate the functionality of different components within the VR system. The test cases have a scenario and expected result and actual result to determine the performance of the system. All the key capabilities, e.g. entering the virtual reality world, changing modules,interacting with kidney models, observing the nephron structures, turning on animations, and studying disease visualizations. To ensure a full system validation, both successful and failure situations are considered. These test cases help to discover issues, justifying requirements, and making sure that all features are up to the needed performance and usability criteria. Each scenario will be described with detail test case tables. and its consequent results.

6.7.1 Test Case 1:Enter VR Environment Successful

This test will ensure that the VR environment loads and works properly once the headset is loaded, properly linked and the program is started. The user is able to enter the VR scene without mistakes or interruptions.

Table 6.1: Test Case:Enter VR Environment Successful

Test Case ID	TC-01
Test Case Name	Enter VR Environment Successful
Description	Verify that the user can successfully enter the VR environment
Preconditions	VR headset connected and application launched
Steps 2. Wear headset 3. Enter VR scene	1. Start application
Expected Result	VR environment loads successfully without errors
Actual Result	VR environment loaded correctly
Status	Pass

6.7.2 Test Case 2:Enter VR Environment Failed

This test validates the system's response when the VR device is not properly set up before launching the application. The system rightly stops the environment loading,ensuring correct error management.

Table 6.2: Test Case:Enter VR Environment Failed

Test Case ID	TC-02
Test Case Name	Enter VR Environment Failed
Description	Verify system behavior when VR initialization fails
Preconditions	VR device not properly connected
Steps	1. Launch application without proper VR setup
Expected Result	System shows error or does not proceed
Actual Result	VR environment did not load
Status	Fail

6.7.3 Test Case 3:Show Kidney Successful

This test confirms that the kidney 3D model appears in the right way in the VR environment upon user interaction. The model was presented in the right form, which validated that asset pipeline is performing as desired.

Table 6.3: Test Case:Show Kidney Successful

Test Case ID	TC-03
Test Case Name	Kidney Visualization Successful
Description	Verify kidney model loads and displays correctly
Preconditions	User inside VR environment
Steps	1. Click "Show Kidney" button
Expected Result	Kidney model appears with correct structure
Actual Result	Model displayed correctly
Status	Pass

6.7.4 Test Case 4:Show Kidney Failed

This test is used to test system behavior in the event that if the kidney model file is corrupted or lost. The system responded in the right way by not showing the model and showed elegant management of lost assets.

Table 6.4: Test Case:Show Kidney Failed

Test Case ID	TC-04
Test Case Name	Kidney Visualization Failed
Description	Verify behavior when kidney model fails to load
Preconditions	Missing or corrupted model file
Steps	1. Click "Show Kidney"
Expected Result	Error or no display
Actual Result	Model did not load
Status	Fail

6.7.5 Test Case 5:Show Nephron Successful

This test guarantees that the nephron model loads and show when triggered within the active VR space. This model was rendered successfully and proved the visualization module works as expected.

Table 6.5: Test Case:Nephron Visualization Successful

Test Case ID	TC-05
Test Case Name	Nephron Visualization Successful
Description	Verify nephron model is displayed correctly
Preconditions	VR environment active
Steps	1. Click "Show Nephron"
Expected Result	Nephron model loads correctly
Actual Result	Model displayed successfully
Status	Pass

6.7.6 Test Case 6:Show Nephron Failed

The test is used to test how the system responds in the case when the nephron model asset is not available. The system gave no output of display as it is known to deal with missing model files without crashing.

Table 6.6: Test Case:Nephron Visualization Failed

Test Case ID	TC-06
Test Case Name	Nephron Visualization Failed
Description	Verify behavior when nephron fails to load
Preconditions	Model asset missing
Steps	1. Click "Show Nephron"
Expected Result	No response
Actual Result	Model not displayed
Status	Fail

6.7.7 Test Case 7: View Kidney Labels Successful

This test confirms that, using the VR controller, if user grabs the part of the kidney in the correct manner it shows lable of the parts. The interaction works well and improves the informational aspect of the VR.

Table 6.7: Test Case:View Kidney Labels Successful

Test Case ID	TC-05
Test Case Name	View Kidney Labels Successful
Description	Verify that grabbing a component separates the mesh and shows labels.
Preconditions	Kidney model is visible and loaded in the scene.
Steps	1. Grab a kidney component using the VR controller.
Expected Result	Internal meshes separate and labels/descriptions appear.
Actual Result	Meshes separated and labels displayed as expected.
Status	Pass

6.7.8 Test Case 8: View Kidney Labels Failed

This test is a test of behavior in the case of mesh separation failure when interaction of the controller occurs.Instead of showing labels the system reloaded the kidney model, showing there is backup mechanism.

Table 6.8: Test Case:View Kidney Labels Failed

Test Case ID	TC-06
Test Case Name	View Kidney Labels Failed
Description	Verify system behavior when mesh separation fails upon interaction.
Preconditions	Kidney model is visible and loaded in the scene.
Steps	1. Attempt to grab a kidney component to trigger separation.
Expected Result	System separates meshes and displays corresponding labels.
Actual Result	Mesh separation failed; system reloaded the kidney model.
Status	Fail

6.7.9 Test Case 9: Kidney Function Animation Successful

This test proves the animation of kidney functionality works smoothly when activated by the user. The animation module was accurate giving a good visual modeling of kidney functioning.

Table 6.9: Test Case:Kidney Animation Successful

Test Case ID	TC-09
Test Case Name	Kidney Animation Successful
Description	Verify kidney function animation plays correctly
Preconditions	Animation module loaded
Steps	1. Click “Kidney Animation”
Expected Result	Animation runs smoothly
Actual Result	Animation played correctly
Status	Pass

6.7.10 Test Case 10:Kidney Function Animation Failed

This test is used to test the behavior of the system in the absence of the animation component or in case of any unavailability. There was no animation shown, confirming that the system manages missing assets related to animation without contributing to system-wide breakdown.

Table 6.10: Test Case:Kidney Animation Failed

Test Case ID	TC-10
Test Case Name	Kidney Animation Failed
Description	Verify behavior when animation fails
Preconditions	Animation component missing
Steps	1. Trigger animation
Expected Result	Animation does not play
Actual Result	No animation shown
Status	Fail

6.7.11 Test Case 11:Kidney Disease Visualization Successful

It is used to confirm the models of kidney diseases that is PKD and kidney stones plays the simulations correctly when selected. The models came out anticipated,checking the feature of visualizing the disease is working properly.

Table 6.11: Test Case 11:Disease Visualization Successful

Test Case ID	TC-11
Test Case Name	Disease Visualization Successful
Description	Verify disease models (PKD / Stones) display correctly
Preconditions	Disease module active
Steps	1. Click “Show Diseases”
Expected Result	Disease models appear correctly
Actual Result	Models displayed correctly
Status	Pass

6.7.12 Test Case 12:Kidney Disease Visualization Failed

This test checks the behavior of the system in case of missing disease model assets. The system did not display any output, which confirms that it is capable of using missing assets without impacting overall application stability.

Table 6.12: Test Case:Disease Visualization Failed

Test Case ID	TC-12
Test Case Name	Disease Visualization Failed
Description	Verify behavior when disease models fail
Preconditions	Missing assets
Steps	1. Click “Show Diseases”
Expected Result	No display
Actual Result	Model not shown
Status	Fail

6.7.13 Test Case 13:Navigation Menu Successful

This test ensures that the VR navigation menu loads the corresponding module correctly when user presses different buttons. The navigation was good,assuring trustworthy UI interaction in the VR environment.

Table 6.13: Test Case 13:Navigation Menu Successful

Test Case ID	TC-13
Test Case Name	Navigation Menu Successful
Description	Verify navigation between modules
Preconditions	VR environment active
Steps	1. Click different buttons
Expected Result	Correct module loads
Actual Result	Navigation worked correctly
Status	Pass

6.7.14 Test Case 14:Navigation Menu Failed

This test ensures that if the VR navigation menu does not load due to an error the corresponding module in this case navigation bar will not appear. The confirms that system detects issue and limits user interaction when menu is not available.

Table 6.14: Test Case:Navigation Menu Failed

Test Case ID	TC-14
Test Case Name	Navigation Menu Failed
Description	Verify behavior when navigation fails
Preconditions	UI script error
Steps	1. Click buttons
Expected Result	No response or wrong module
Actual Result	Navigation failed
Status	Fail

6.8 Summary

This chapter described the testing and assessment procedures used to verify the reliability, performance, and usefulness of the VR Anatomy Learning system. Various testing approaches, including GUI testing, usability testing, performance testing, compatibility testing, and exception handling, were used to validate various components of the system. The findings show that the program works well and offers a seamless and flexible learning experience. The system has been continuously tested and refined to fulfill both technological and educational standards, assuring its efficacy as a virtual learning tool.

CHAPTER 7
CONCLUSION

7.1 Conclusion

The project titled *VR Anatomy Learning: Kidney Visualization* is an innovative approach to medical education by integrating virtual reality technology with interactive 3D anatomical models. The system successfully addresses the constraints of conventional learning techniques, like textbooks and fixed diagrams, through offering an immersive environment that allows users to learn about the structure and functionality of the human kidney in a more captivating, intuitive way. By using Unity as well as the XR Interaction Toolkit, the app will allow users to interact with detailed 3D models, view internal structures by separating meshes and understand physiological processes through animations and visual simulations.

The system was designed with a focus on usability, performance, and educational effectiveness throughout the development process.Implementation of various modules,which includes kidney visualization, nephron exploration, disease representation, and functional animations, guarantees a comprehensive learning experience. The integration of user interaction such as grabbing, selecting increase the engagement and allows learners to be actively participate in the learning and not a passive consumer of information.Moreover,the disease models like Polycystic Kidney Disease and kidney stones are included that offers valuable information on a real-world medical conditions,further strengthening the educational value of the system

Testing and evaluation phase ensured that the system is effective and efficient and achieves the desired goals. Several testing methods, such as GUI testing, usability testing, performance testing and compatibility testing, were performed to ensure reliability and smooth operation.The results illustrated that the application offers reliable performance, responsive interactions, and easy-to-use user-friendly interface. Overall, the project achieves its vision of providing a powerful and interactive anatomy learning tool.

7.2 Limitations of the System

Although the system has been successfully implemented, it has some limitations that can be addressed in future improvements. One of the main limitations is the emphasis on a single organ, the kidney, that limits the scope of the application to the broader anatomical learning. Having more organs in the system would make it more useful in medical education.The dependency on VR hardware is another limitation,which might not be easily available to all users, thus limiting its widespread adoption.

Additionally, the system now offers predefined interactions and animations, which can limit the amount of customization available to the user.Some users can also get motion sickness or will need time to get used to the VR environment. Furthermore, the level of anatomical detail, while sufficient for educational purposes,can be further improved to match advanced medical standards. Addressing these limitations would greatly enhance the flexibility, accessibility, and educational effectiveness of the system.

7.3 Future Enhancements

The proposed system offers solid foundation for further development and expansion.More anatomical modules can be integrated in future to include other organs and body systems, which will convert the application into a full-scale anatomy learning platform. Quizzes and assessment modules, as well as progress tracking can be added with an aim to track student learning outcomes and delivering personalized feedback.

The use of augmented reality (AR) feature is another possible improvement to enhance the VR experience and enable the system to be used on mobile devices.More interactive mechanisms and intelligent learning features can also be implemented to improve the system and to provide a more adaptive, engaging and user-centered educational experience.

APPENDIX A

USER MANUAL

User Manual provides detailed instructions on the usage of the Virtual Reality Kidney Learning System (VR). This manual covers the system requirements,how users may launch the application, and how to communicate with different features like kidney visualization, nephron exploration, animations, anatomical positioning and disease visualization. It is made to assist students in navigating effectively and utilize the system for an immersive learning experience.

A.1 System Requirements

The VR Kidney Learning System requires the following components to operate:

- **Device:** VR-ready PC or laptop with sufficient graphics capability.
- **VR Equipment:** Compatible VR headset.
- **Software:** Installed VR application developed in Unity.
- **Input Devices:** VR controllers for interaction.

A.2 Getting Started

A.2.1 Launching the Application

To begin using the system, the user must first launch the VR application by connecting and wearing the VR headset and running the application on the system. Once the application is executed, the virtual environment is automatically started and loaded for the user.

A.3 Using the Application

A.3.1 Entering the VR Environment

Once the application is launched, the user enters the virtual learning environment where the main interface is loaded by the system. A navigation menu with interactive buttons is displayed where user can access various aspects of the system.

A.3.2 Viewing the Kidney

To visualize the kidney model, the user selects the appropriate option on the interface.The kidney model appears in front of the user when the Show Kidney button is pressed,allowing one to interact with it and closely examine its structure.

Figure A.1: VR Environment

Figure A.2: Kidney Visualization

A.3.3 Exploring Kidney Internal Structures

The system allows users to get a closer look at the internal structures of the kidney. Users can use interaction controls to separate the kidney meshes to see the labels and descriptions of individual parts, to understand them better.

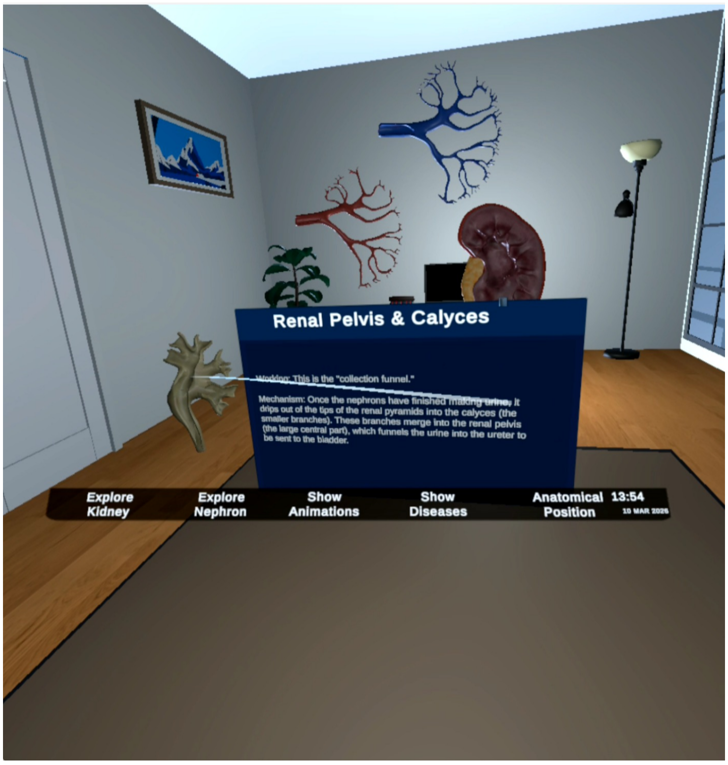

Figure A.3: Kidney Internal Structure Exploration

A.3.4 Viewing the Nephron

Users may view the nephron after exploring the kidney, which is the functional unit of the kidney. The nephron model can be pressed by the Show Nephron button,allowing one to learn its structure in detail.

A.3.5 Viewing Kidney Function Animation

The system provides an animation feature to illustrate the physiological processes of the kidney. When the user presses the Show Animation button, an animation illustrating filtration and urine formation is played,along with an audio narration which explains each step.

Figure A.4: Nephron Visualization

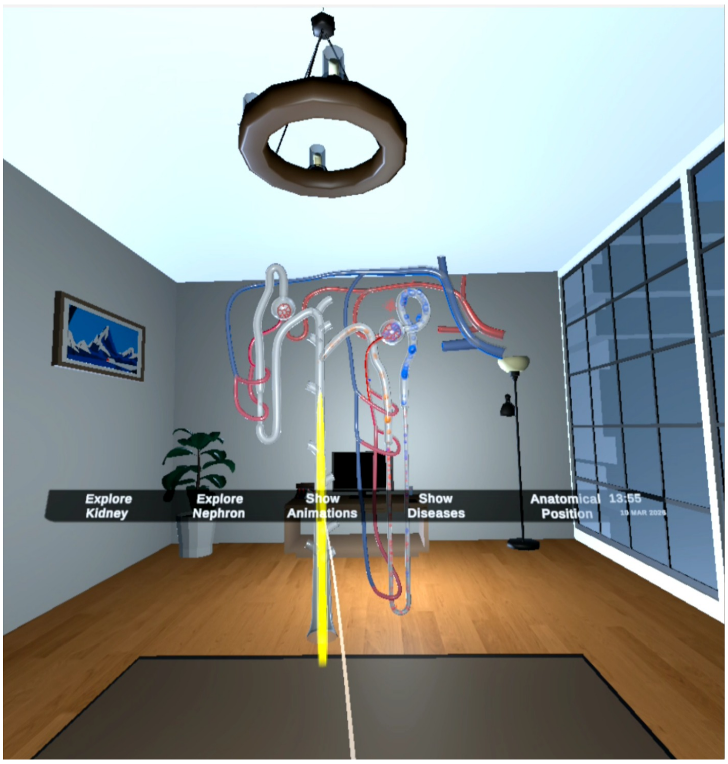

Figure A.5: Kidney Function Animation

A.3.6 Viewing Anatomical Position

It also allows the user to learn where the kidney is located in the human body.When the button of the Anatomical Position is pressed, a complete model of a human body appears and with a short pause, the camera goes into the body to highlight the exact position of the kidney.

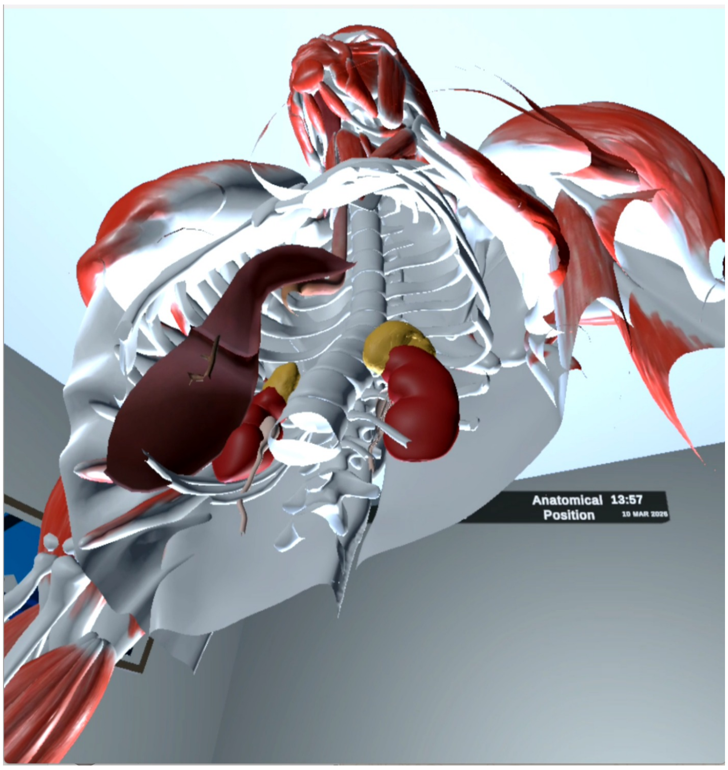

Figure A.6: Anatomical Position View

A.3.7 Viewing Kidney Diseases

The system enables the user to learn about various kidney diseases via interactive visualization. On clicking the Show Diseases button, one will get the opportunity to select either Polycystic Kidney or Kidney Stones. The user is able to choose a disease in order to see the relevant model of an affected kidney, and an audio explanation to understand better.

Figure A.7: Kidney Disease Visualization

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